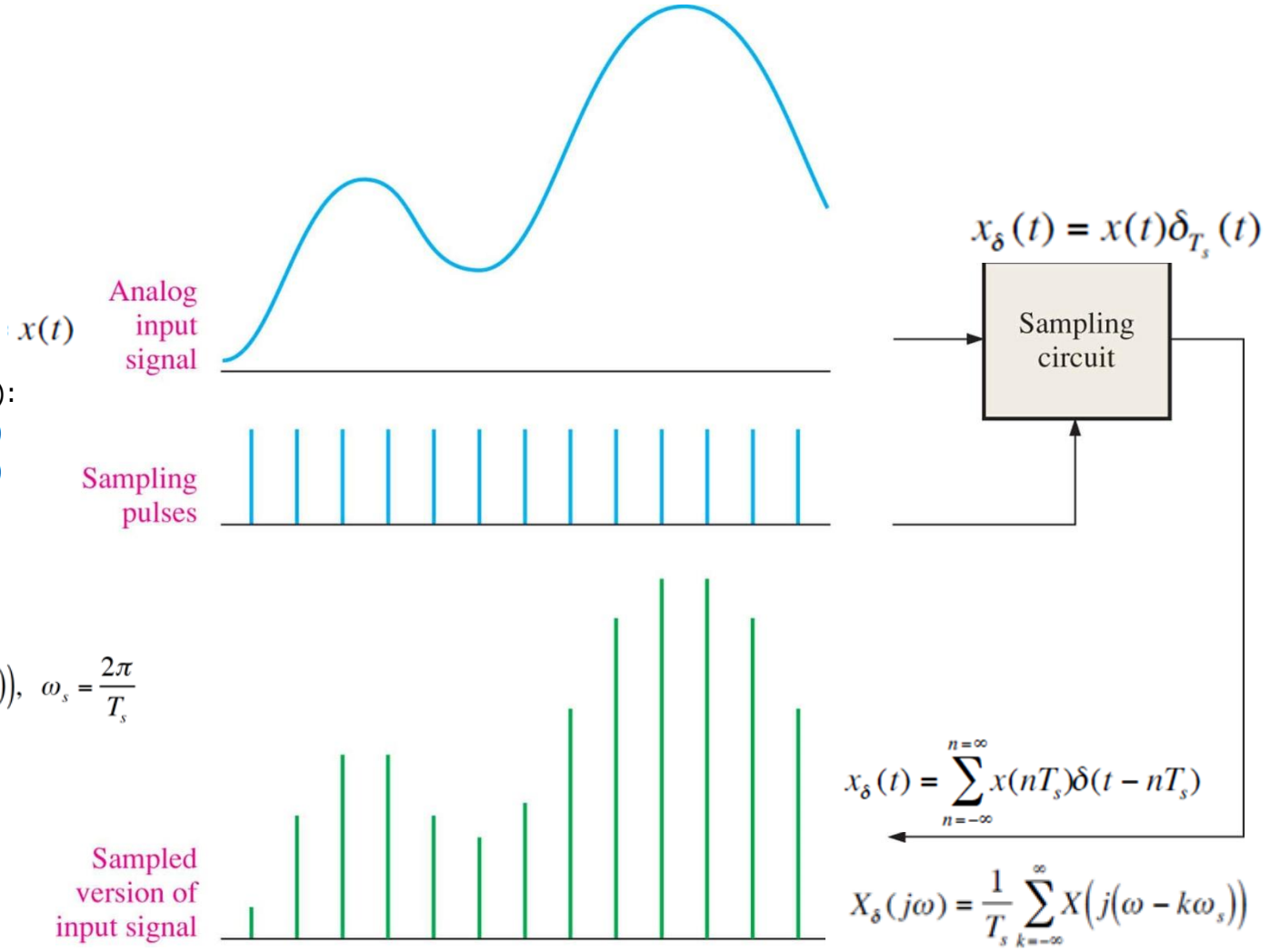
Digital Circuit and Logic Design

Chapter 11

ADC and DAC

Analog-to-Digital Converter

Nyquist Theorem, Sampling process



unit impulse (Discrete time):

$$\delta[m] = \begin{cases} 1 & , \quad m = 0 \\ 0 & , \quad m \neq 0 \end{cases}$$

unit impulse train:

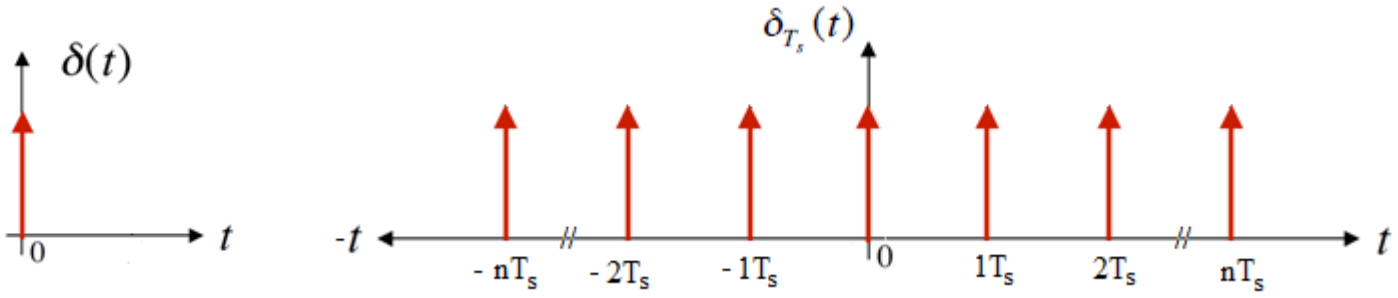
$$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$\delta_{\omega_s}(j\omega) = \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} \delta\left(j(\omega - k\omega_s)\right), \quad \omega_s = \frac{2\pi}{T_s}$$

Discrete-time Signals

A **train of impulses** has Fourier Transform

$$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \longleftrightarrow \delta_{\omega_s}(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(j\left(\omega - k\omega_s\right)\right), \quad \omega_s = \frac{2\pi}{T_s}$$



So **sampling** a continuous-time signal **in the time domain**

$$x_\delta(t) = x(t)\delta_{T_s}(t) = \sum_{n=-\infty}^{n=\infty} x(nT_s)\delta(t - nT_s)$$

is a **convolution in the frequency domain**

$$X_\delta(j\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X\left(j\left(\omega - \sigma\right)\right) \delta_{\omega_s}\left(j\sigma\right) d\sigma$$

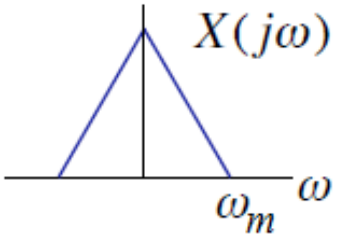
$$= \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X\left(j\left(\omega - k\omega_s\right)\right)$$

Sampling in the Frequency Domain

Graphical interpretation of the formula

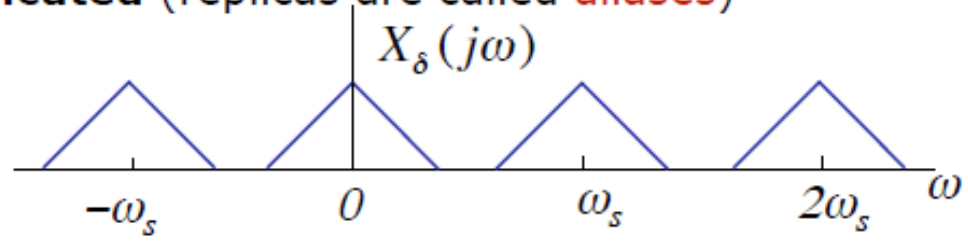
$$X_\delta(j\omega) = \frac{1}{T_s}\sum_{k=-\infty}^{\infty} X\left(j\left(\omega - k\omega_s\right)\right), \quad \omega_s = 2\pi f_s, \quad f_s = \frac{1}{T_s}$$

Continuous-time spectrum



$$\omega_m = 2\pi f_m$$

is **scaled** and **replicated** (replicas are called aliases)



Aliasing: If signal bandwith $f_m \geq \frac{f_s}{2}$ then spectrum overlaps!

The sampling theorem

Suppose a signal's **highest frequency** is f_m (a low-pass or a band-pass signal). Then a proper sampling requires a sampling frequency at least satisfying,

$$f_m < \frac{f_s}{2} = 0.5 \times f_s$$

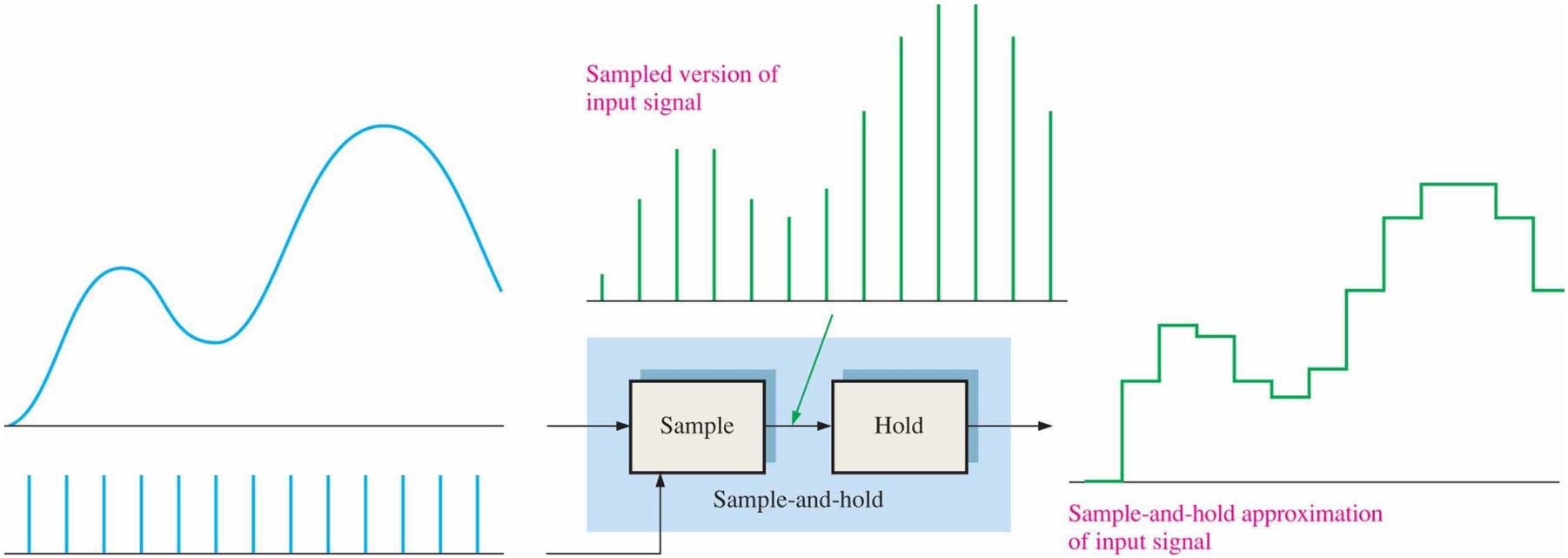
The number $\frac{f_s}{2}$ is called the Nyquist frequency.

The number $2f_m$ is called the Nyquist rate.

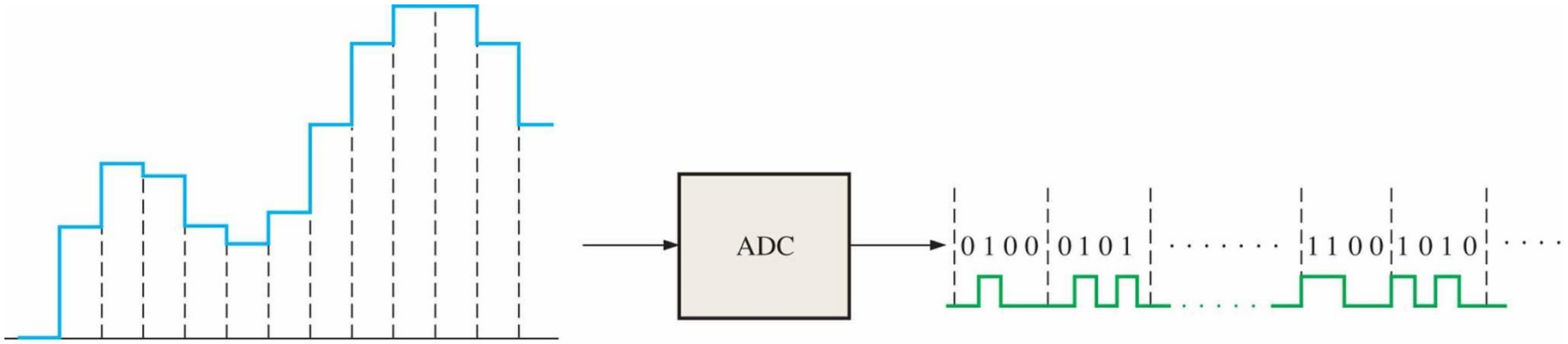
Example: Consider an analog signal with frequencies between 0 and 3kHz. A proper sampling requires a 6kHZ sampling frequency or higher.

Effects of **aliasing**: It can change the signal real frequency and the signal real phase.

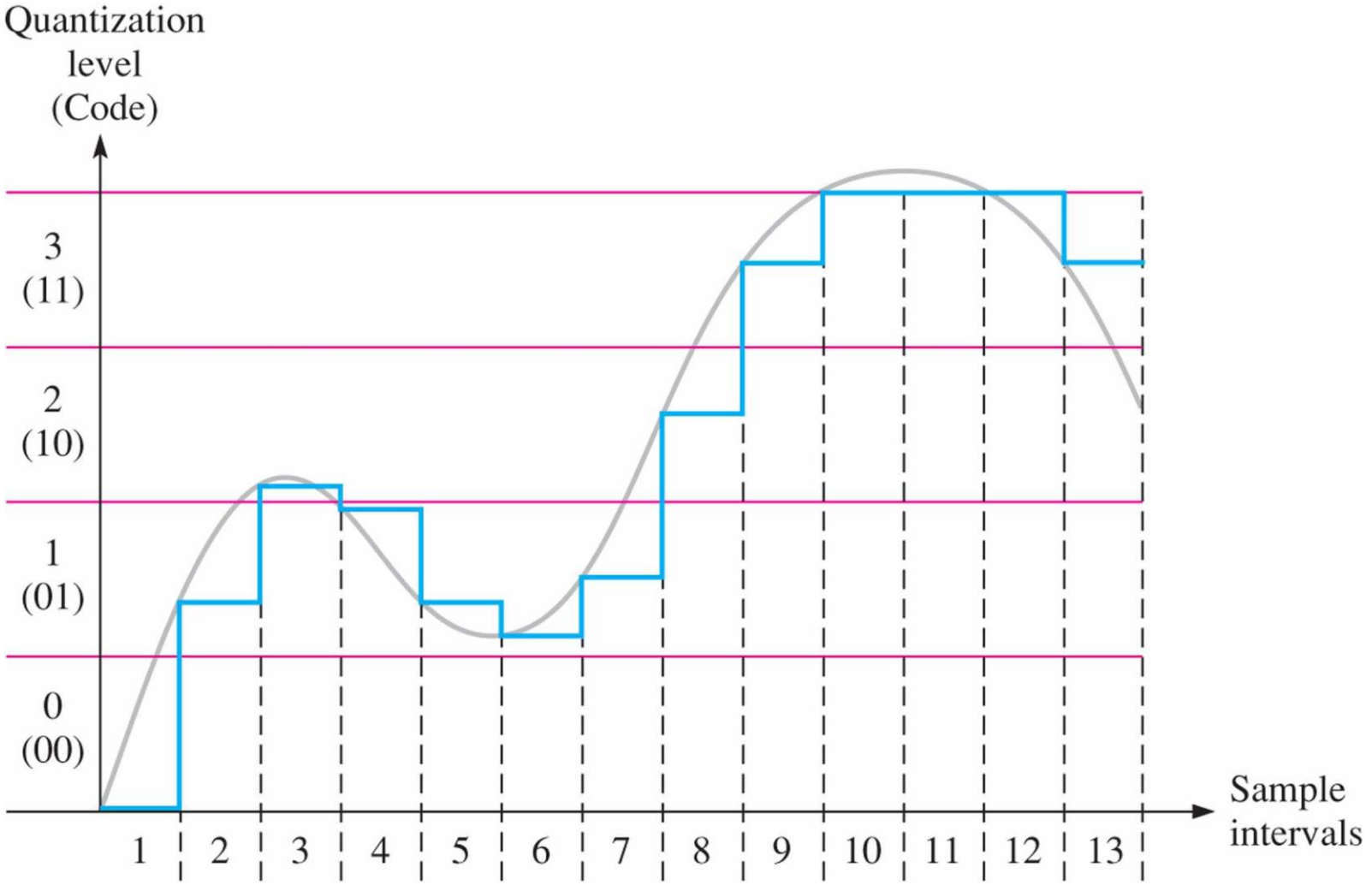
Sample-and-hold operation



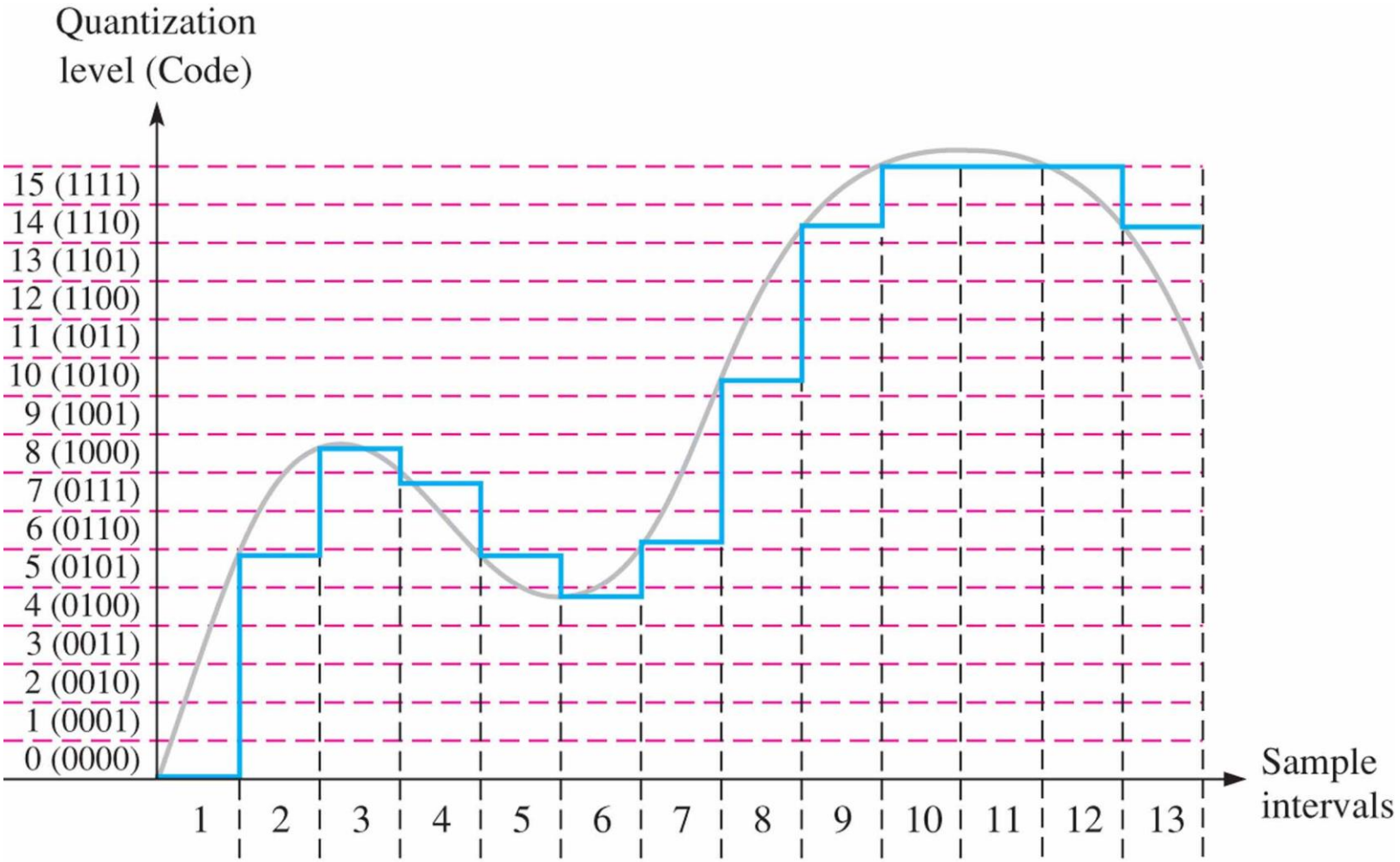
Basic function of an ADC



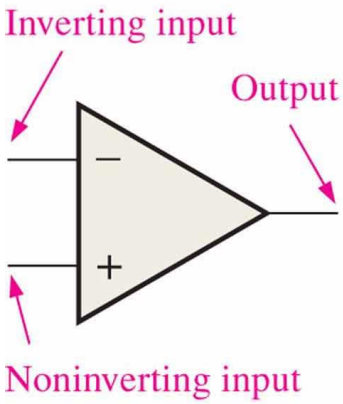
4 quantization levels



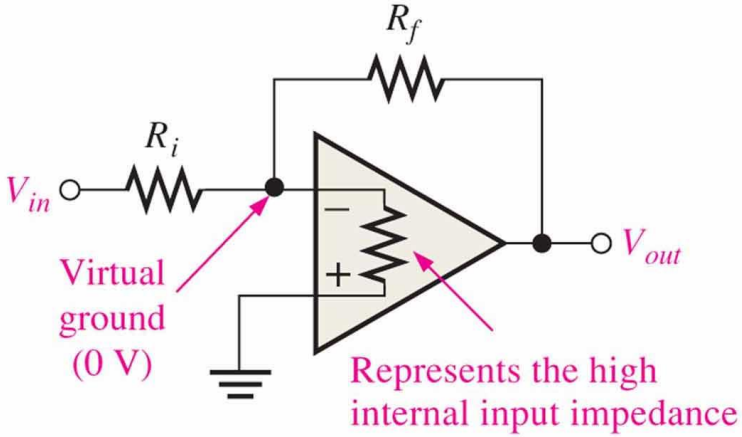
16 quantization levels



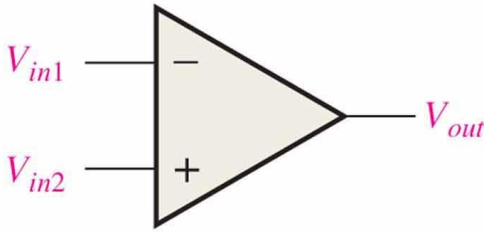
Operational amplifier (op-amp)



(a) Op-amp symbol



(b) Op-amp as an inverting amplifier with gain of R_f/R_i



(c) Op-amp as a comparator

Ideal op-amps

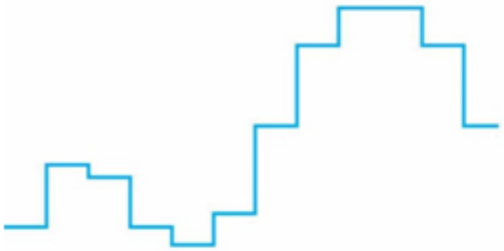
1. Infinite open-loop gain G = v_{out} / v_{in}
2. Infinite input impedance Z_{in}, and so zero input current
3. Zero input offset voltage (V_{in1} = V_{in2})
4. Zero output impedance Z_{out}
5. Infinite output voltage range
6. Infinite bandwidth with zero phase shift and infinite slew rate
7. Zero noise
8. Infinite common-mode rejection ratio (CMRR)
9. Infinite power supply rejection ratio.

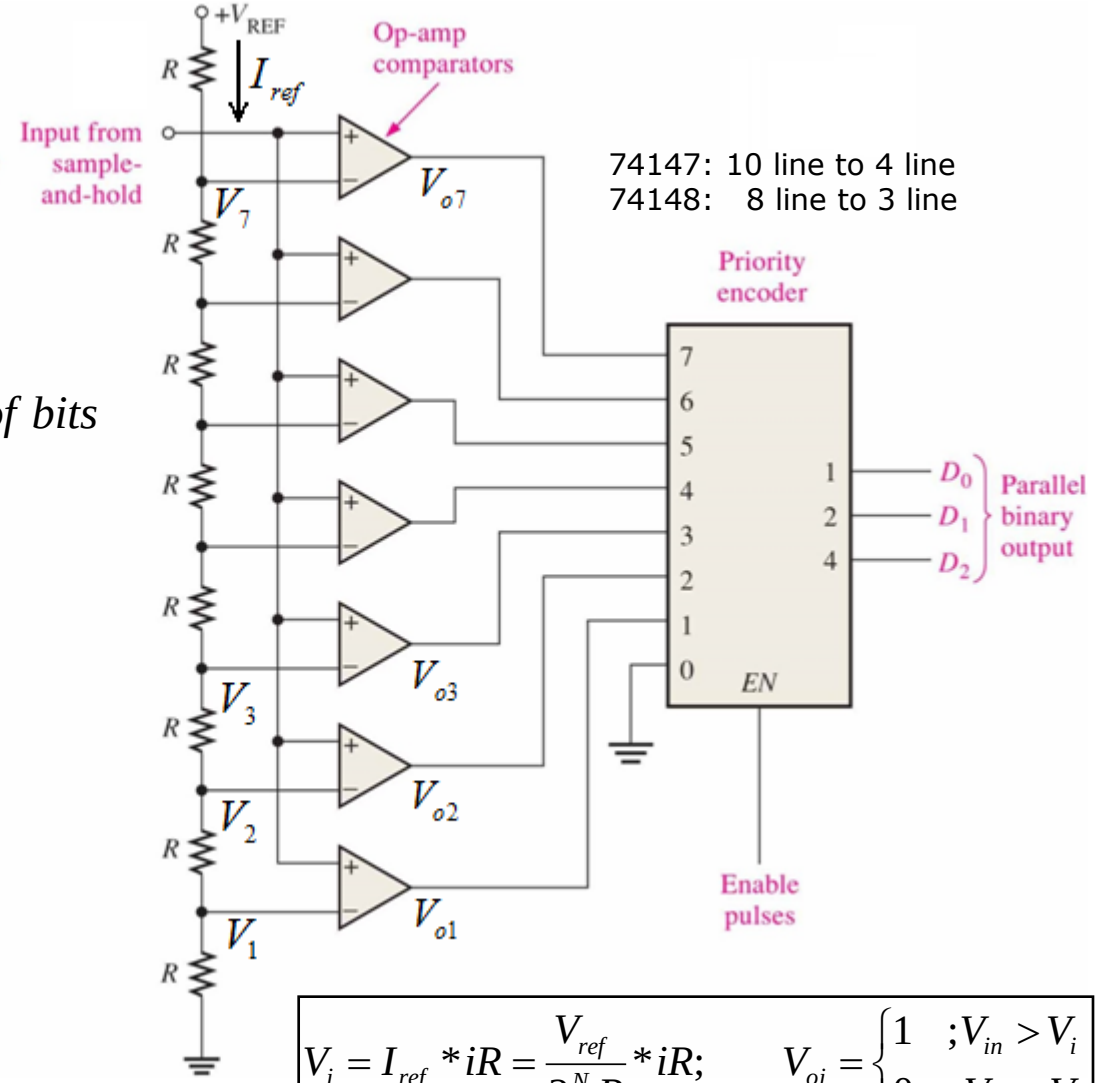
Types of ADCs

Many types of ADCs available. There are 4 types wildly used.

1. Flash ADC
2. Dual-slope ADC
3. Successive-approximation ADC
4. Sigma-delta ADC

A 3-bit flash ADC





$$I_{ref} = \frac{V_{ref}}{\sum R} = \frac{V_{ref}}{2^N R}; \qquad N: number\ of\ bits$$

$$V_1 = I_{ref} * R = \frac{V_{ref}}{2^N R} * R;$$

$$V_2 = I_{ref} * 2R = \frac{V_{ref}}{2^N R} * 2R;$$

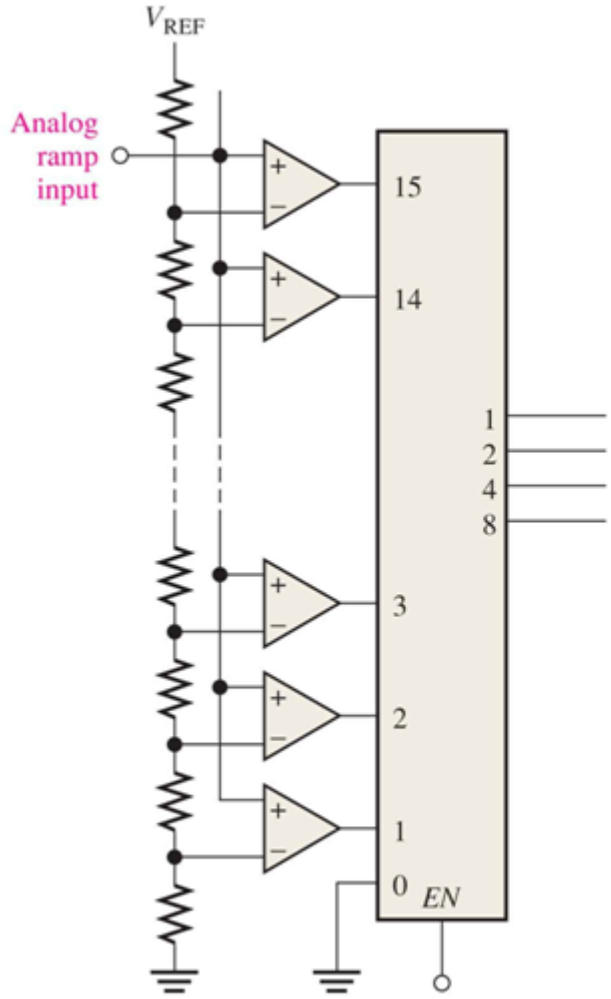
$$V_3 = I_{ref} * 3R = \frac{V_{ref}}{2^N R} * 3R;$$

$$\vdots$$

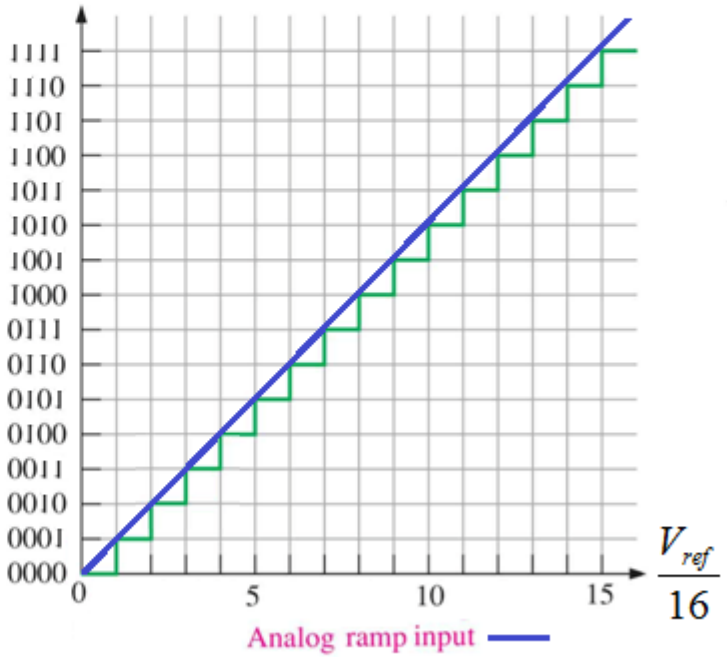
$$V_7 = I_{ref} * 7R = \frac{V_{ref}}{8R} * 7R; \qquad N = 3$$

$$V_i = I_{ref} * iR = \frac{V_{ref}}{2^N R} * iR; \qquad V_{oi} = \begin{cases} 1 & ;V_{in} > V_i \\ 0 & ;V_{in} < V_i \end{cases}$$

A 4-bit flash ADC

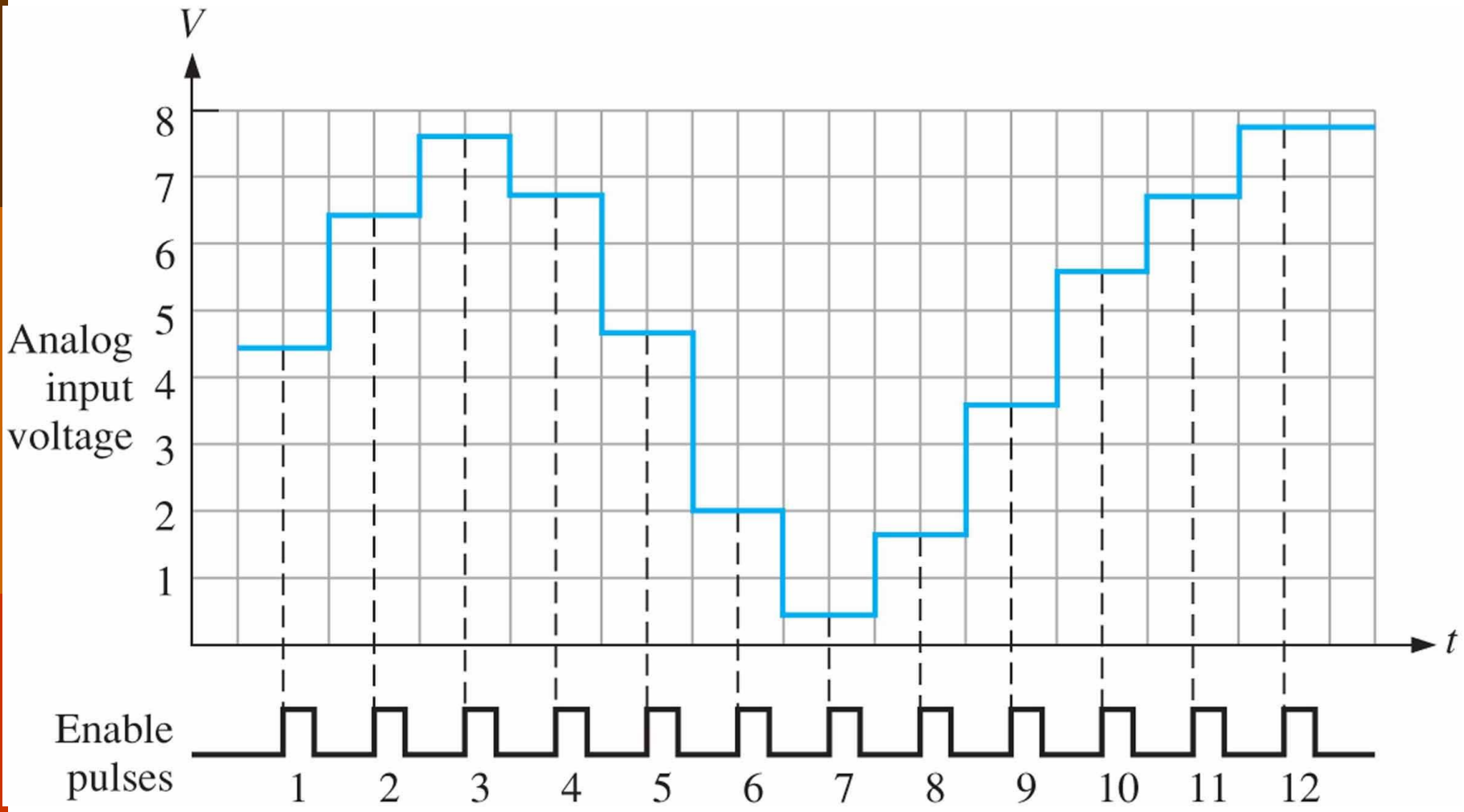


(a)

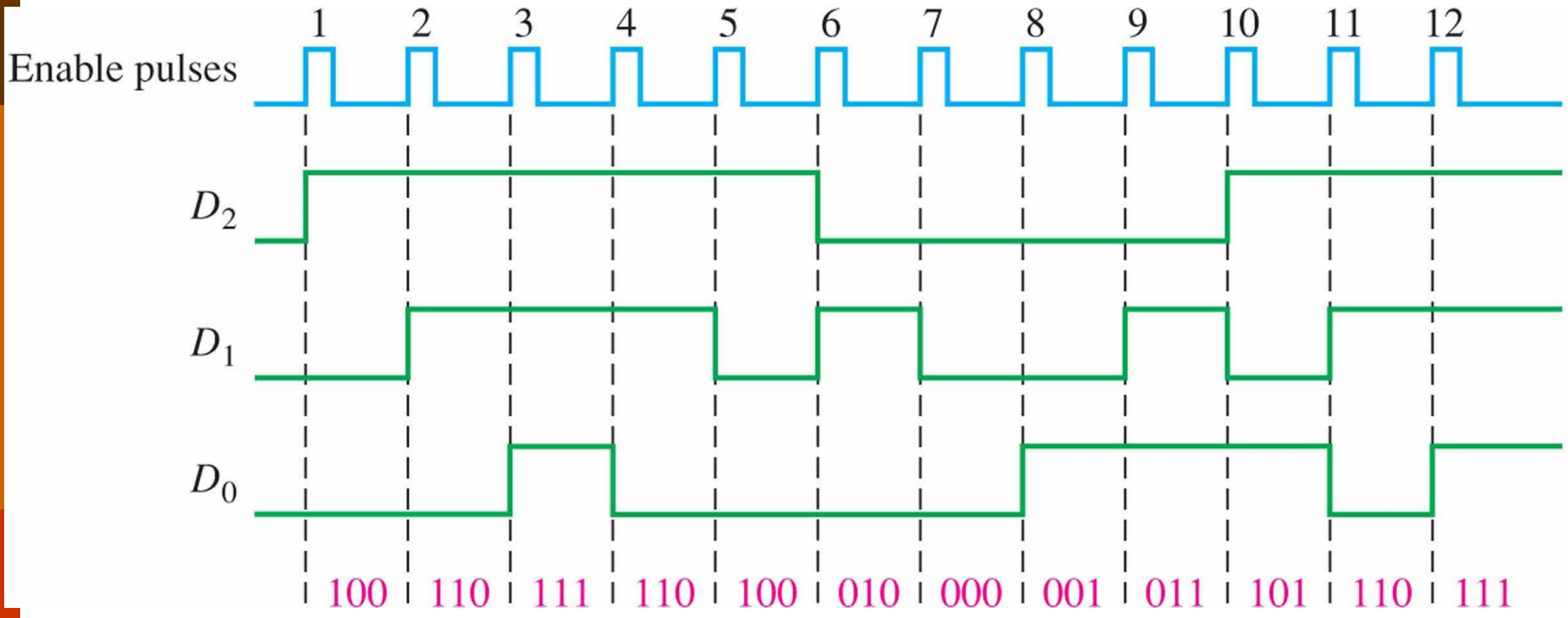


(b)

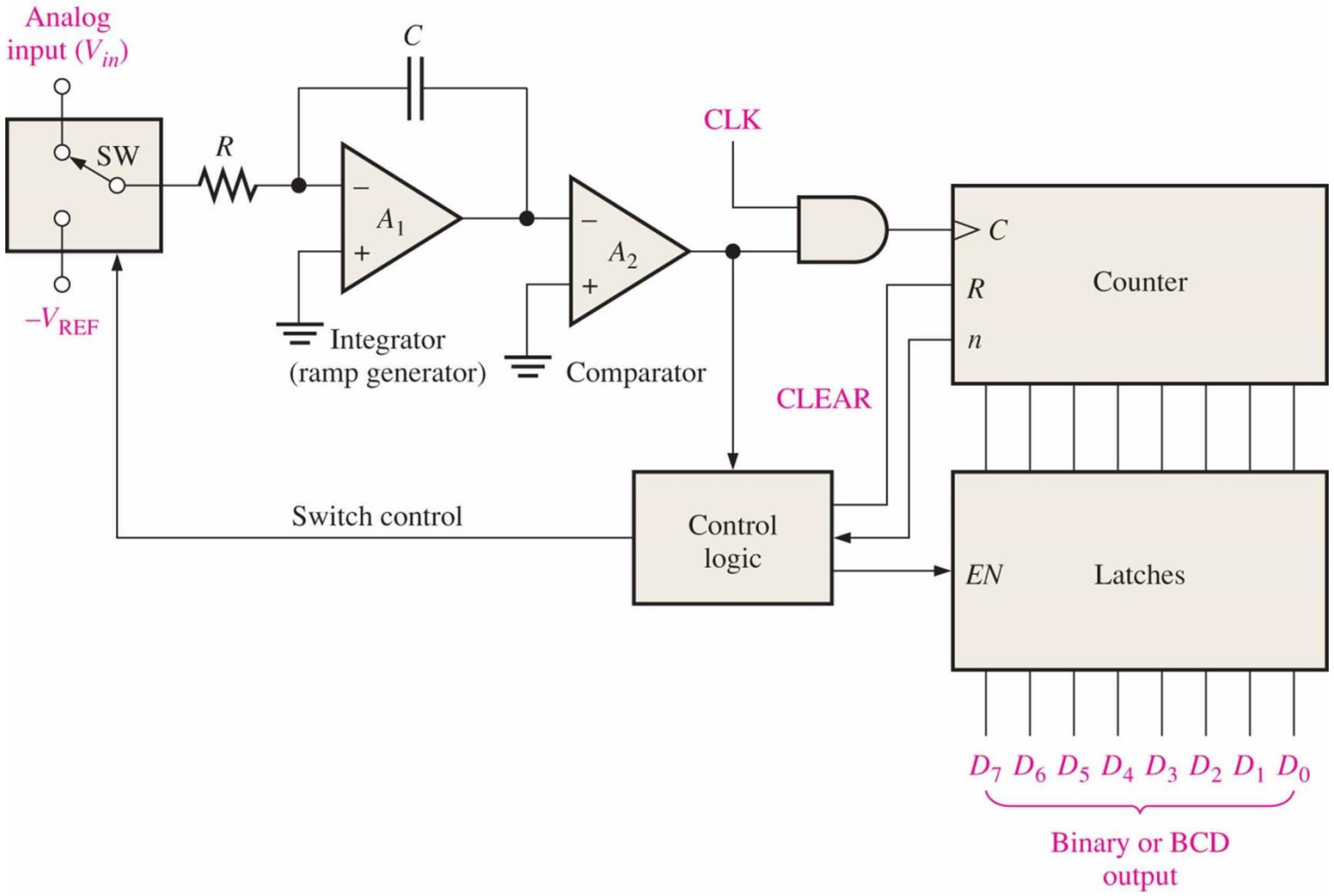
Sampling of values on a waveform for conversion to binary code



Digital outputs for sample-and-hold values



Dual-slope ADC



Dual-slope conversion

Variable slope:

$$I_C = C\frac{dV_C}{dt} \Rightarrow V_C = \frac{1}{C} I_C \Delta t; \quad I_C = \frac{V_{in}}{R};$$

$$V_{C1} = \frac{1}{C}\left(\frac{V_{in}}{R}\right)\Delta t; \quad \Delta t = NT$$

$$V_{C1} = \left(\frac{V_{in}}{RC}\right) * NT; \quad \textit{Variable slope} = \frac{V_{in}}{RC}$$

Fixed slope:

$$I_C = C\frac{dV_C}{dt} \Rightarrow V_C = \frac{1}{C} I_C \Delta t; \quad I_C = \frac{V_{ref}}{R};$$

$$V_{C2} = \frac{1}{C}\left(\frac{V_{ref}}{R}\right)\Delta t; \quad \Delta t = nT$$

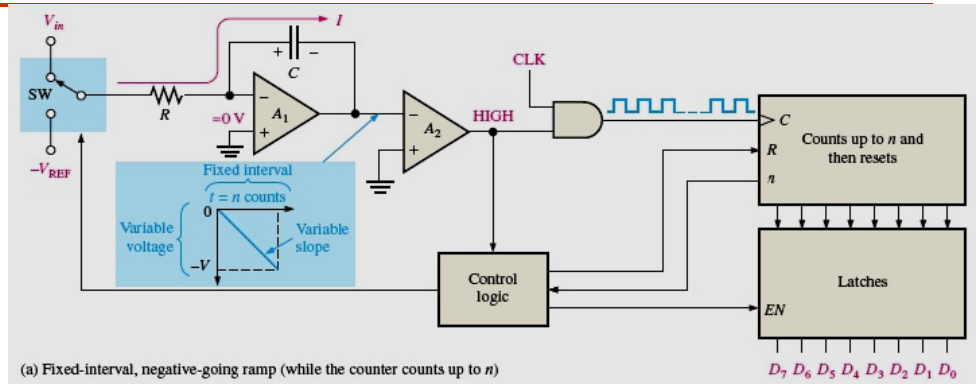
$$V_{C2} = \left(\frac{V_{ref}}{RC}\right) * nT; \quad \textit{Fixed slope} = \frac{V_{ref}}{RC}$$

$$V_{C2} - V_{C1} \geq 0$$

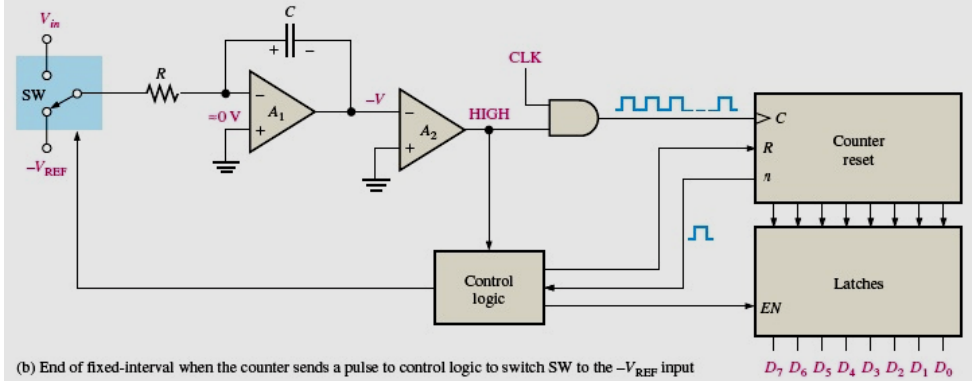
$$\left(\frac{V_{ref}}{RC}\right) * nT - \left(\frac{V_{in}}{RC}\right) * NT = 0$$

$$n = \left(\frac{V_{in}}{RC}\right) * NT * \left(\frac{RC}{V_{ref}}\right)\frac{1}{T}$$

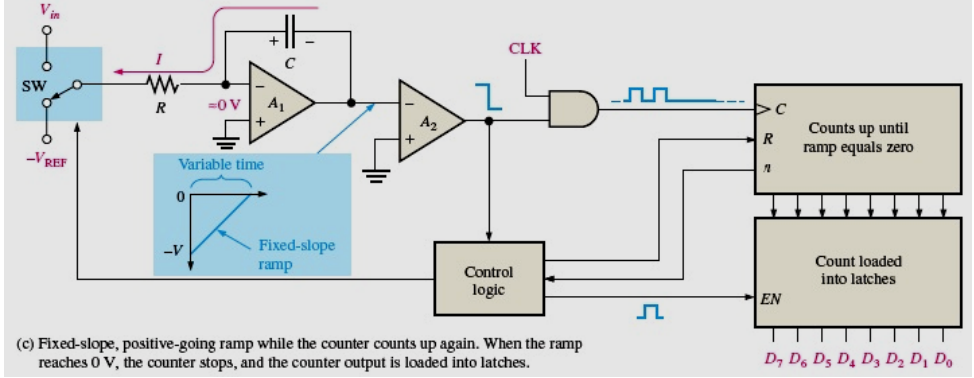
$$n = \left(\frac{N}{V_{ref}}\right) V_{in}$$



(a) Fixed-interval, negative-going ramp (while the counter counts up to n)

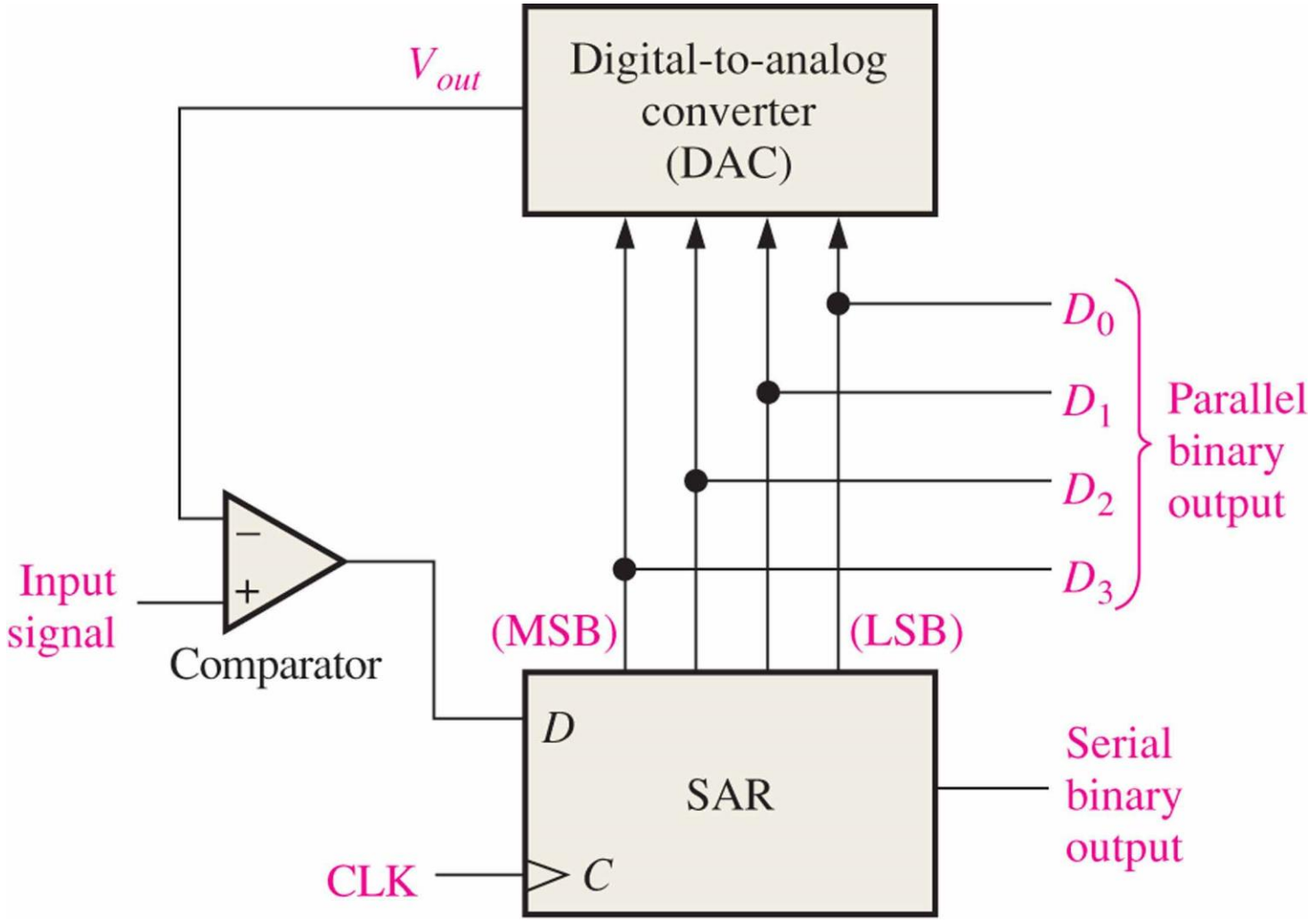


(b) End of fixed-interval when the counter sends a pulse to control logic to switch SW to the $-V_{REF}$ input



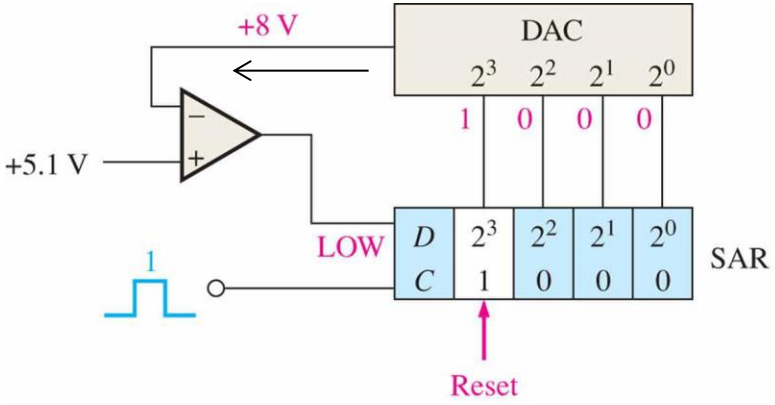
(c) Fixed-slope, positive-going ramp while the counter counts up again. When the ramp reaches 0 V, the counter stops, and the counter output is loaded into latches.

Successive-approximation ADC

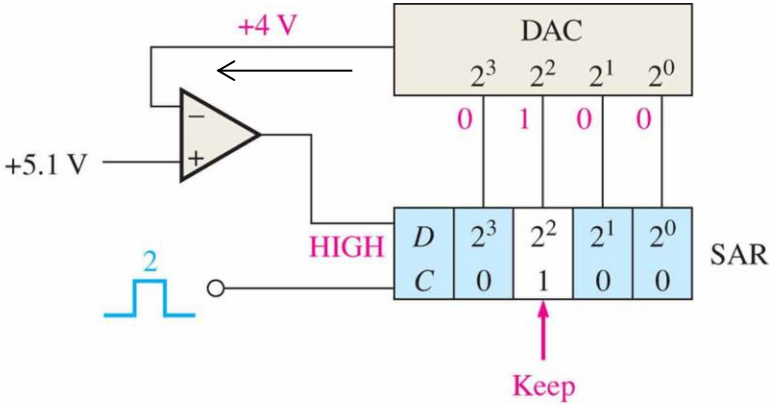


*Successive-approximation register

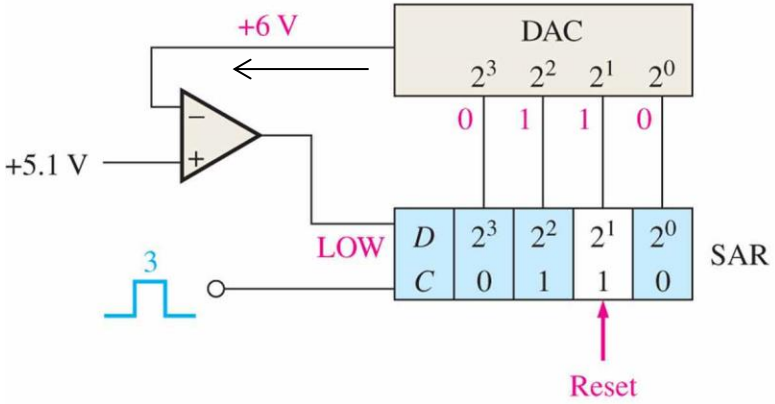
Successive-approximation conversion process



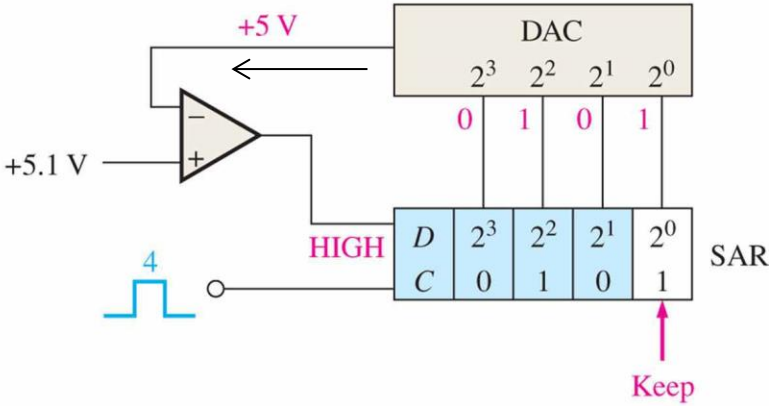
(a) MSB trial



(b) 2^2-bit trial

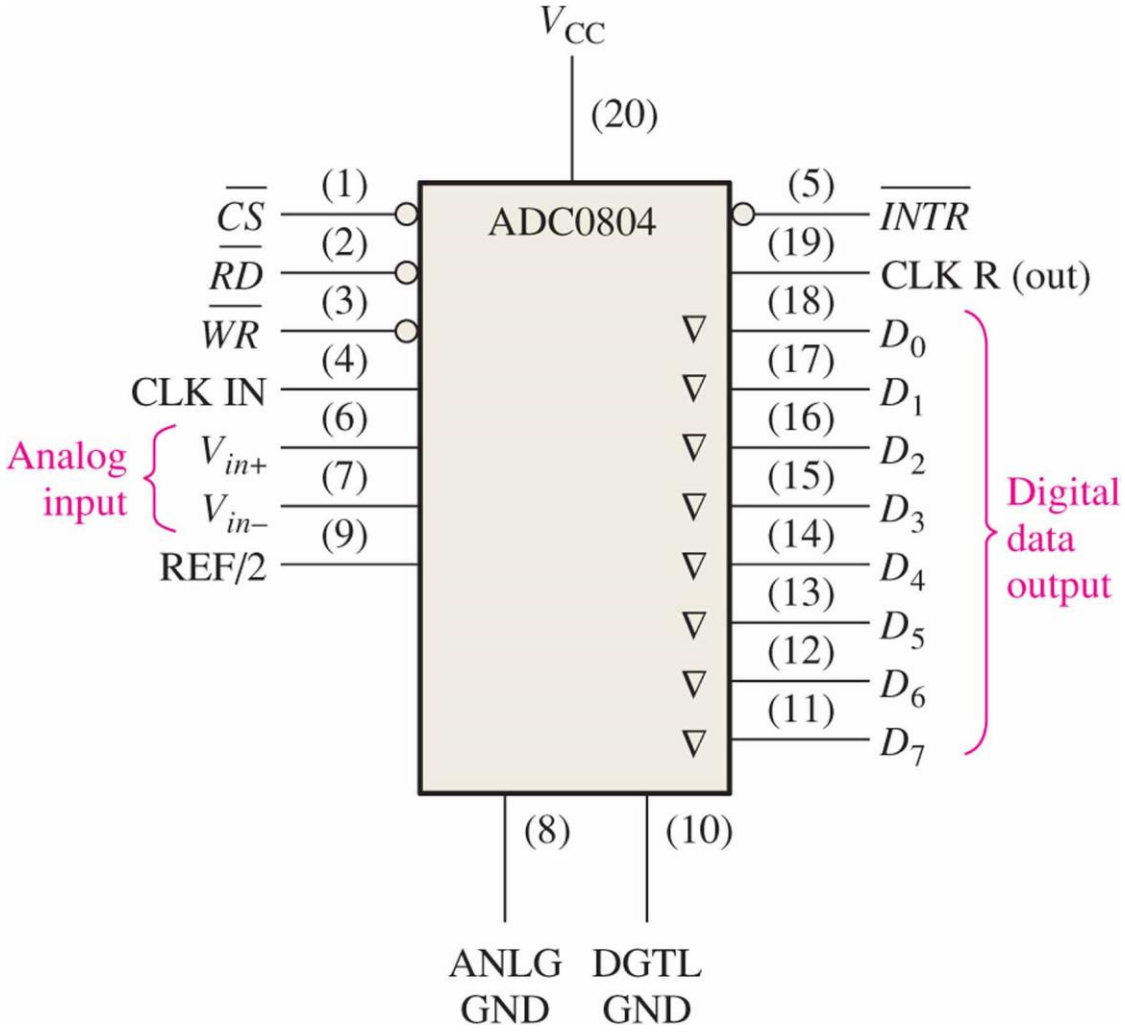


(c) 2^1-bit trial

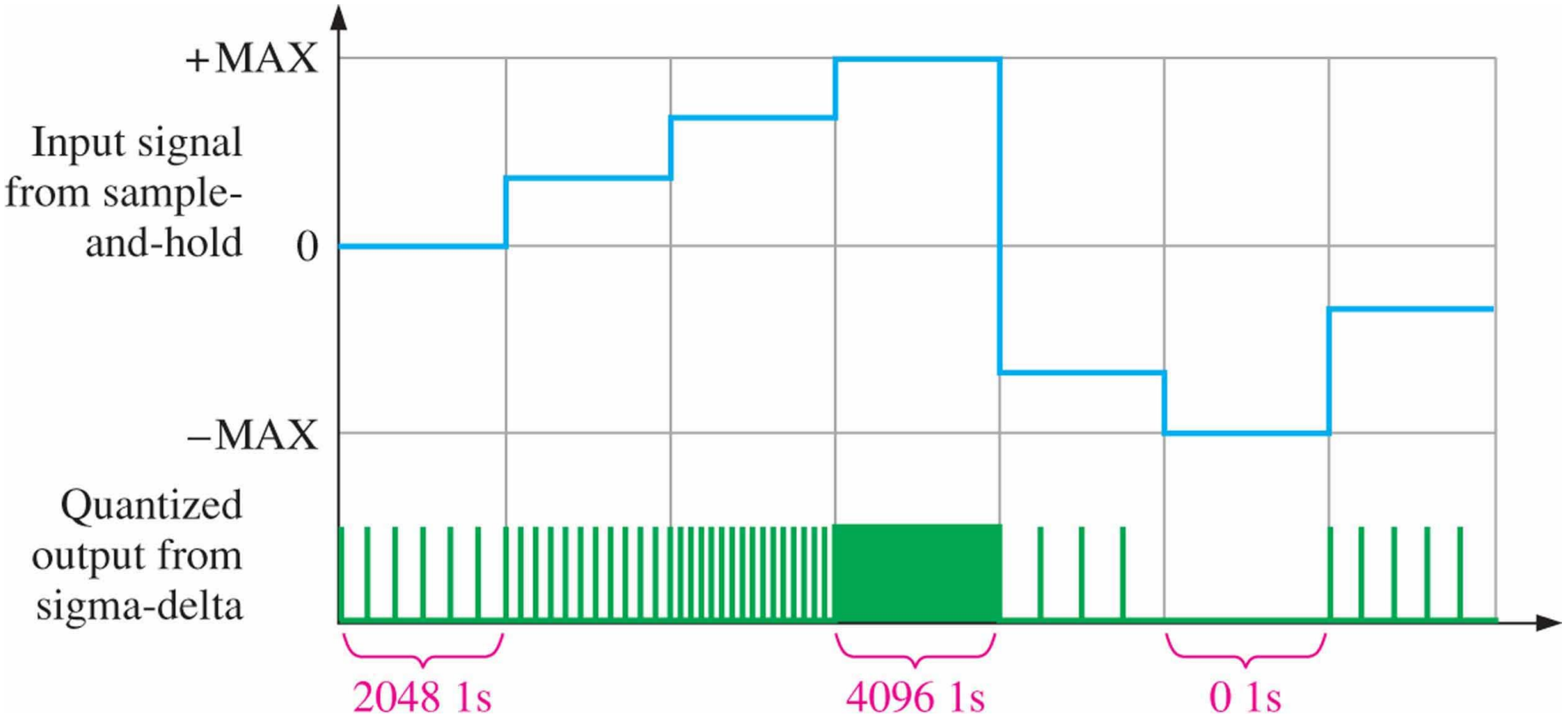


(d) LSB trial (conversion complete)

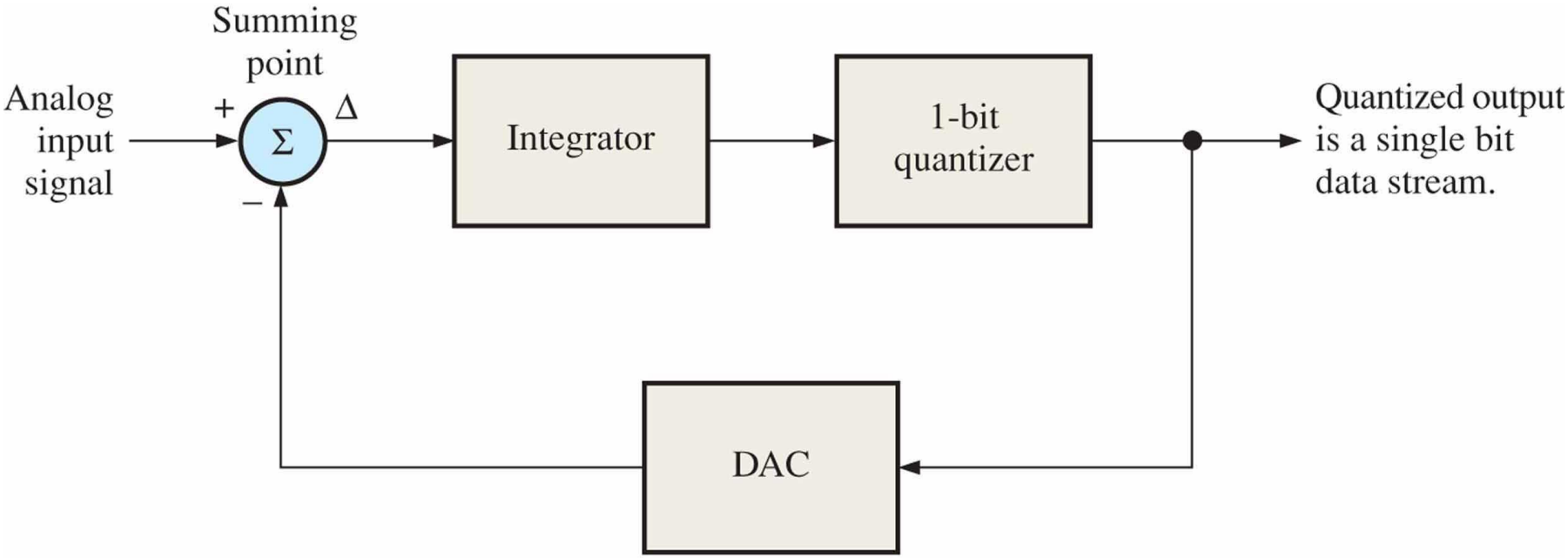
ADC0804 analog-to-digital converter (75.00฿)



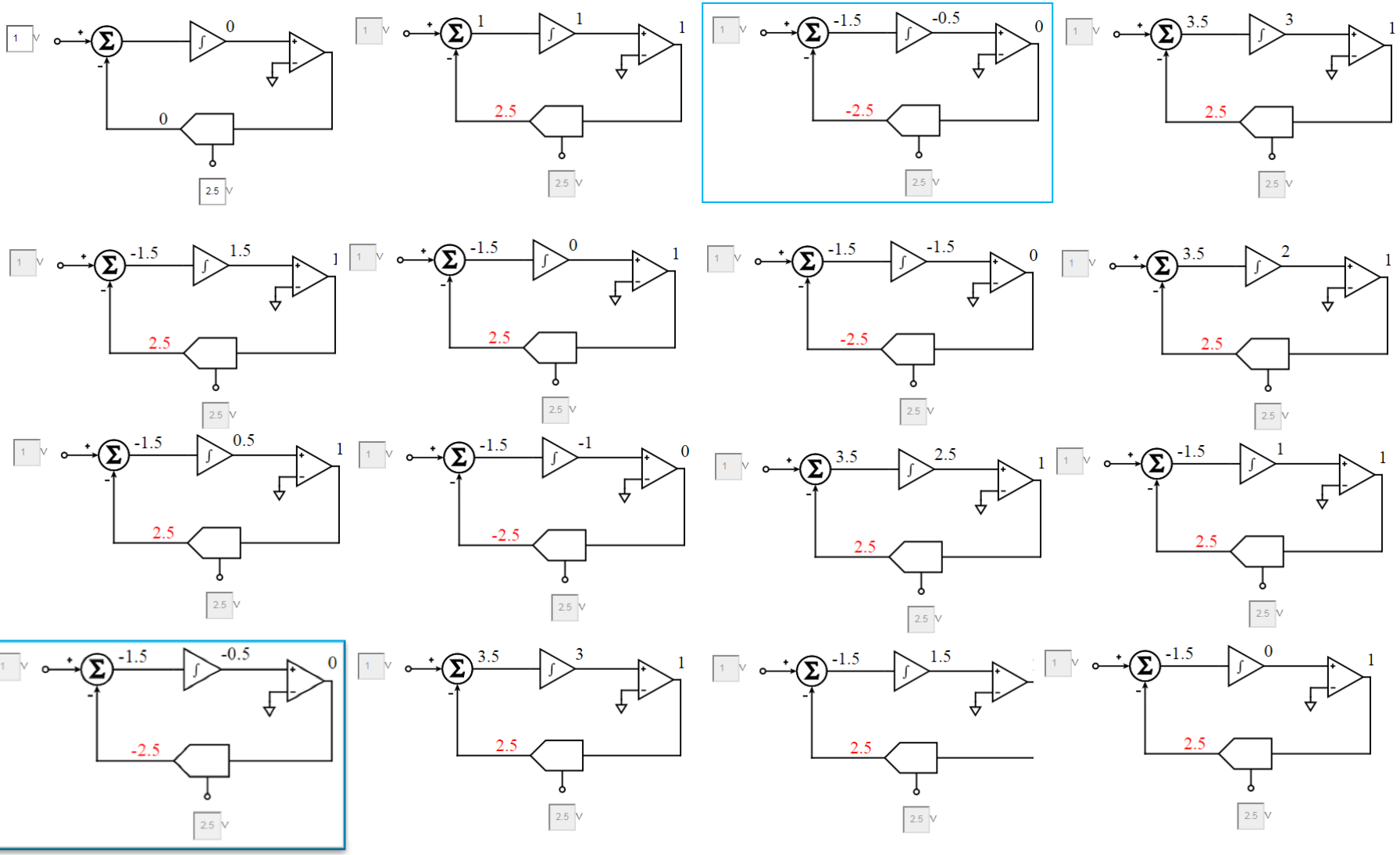
Sigma-delta analog-to-digital conversion



Block diagram of a sigma-delta ADC



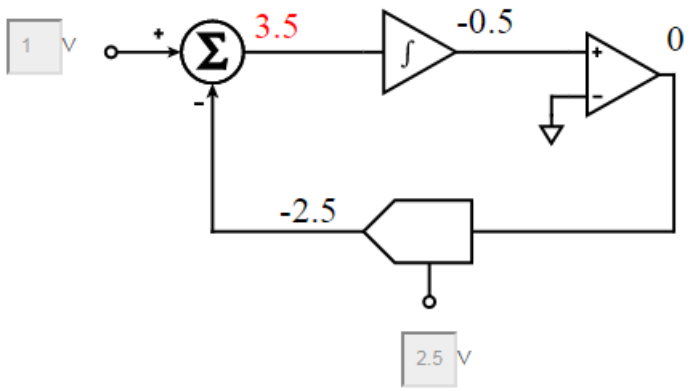
One type of sigma-delta ADC

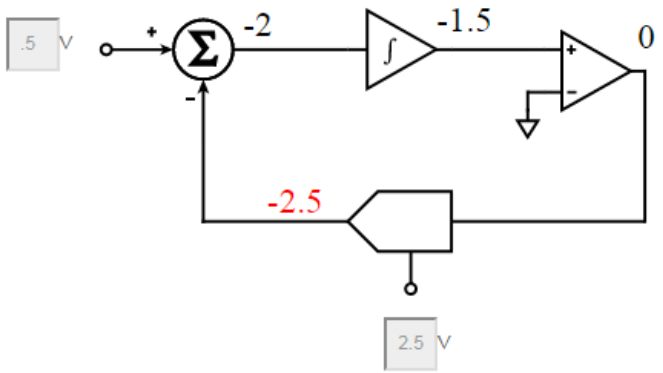


Bit Stream: 1 0 1 1 1 0 1 1 0 1 1 0 1 1 1

http://www.analog.com/en/design-center/interactive-design-tools/sigma-delta-adc-tutorial.html

One type of sigma-delta ADC





Bit Stream: 1 1 1 0 1 1 0 1 1 0 1 1 1 0 1 1 0 1 1 0 1 1 1 0 1 1 0 1 1 0

Mean Output: 0.996

The adder (delta) output becomes 3.5V.

Start Over | Previous Step | Next Step | Next 512 Loops

Bit Stream: 1 0 1 1 0 1 0 1 1 0 1 0 1 1 0 1 0 1 1 0 1 0 1 1 0 1 0 1 1 0

Mean Output: 0.498

The comparator output is 0, so the DAC output becomes -2.5

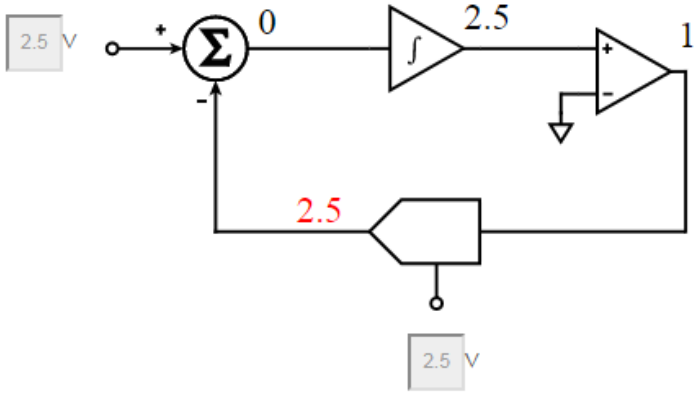
Start Over | Previous Step | Next Step | Next 512 Loops

$$V_{average} = \frac{no.1}{N} * (V_{ref} - (-V_{ref})) - V_{ref}; \quad V_{ref} = 2.5, N = 10(long\ run)$$

$$1 = \frac{7}{10} * (2.5 - (-2.5)) - 2.5; \qquad 2.5 = \frac{10}{10} * (2.5 - (-2.5)) - 2.5$$

$$0 = \frac{5}{10} * (2.5 - (-2.5)) - 2.5; \qquad -2.5 = \frac{0}{10} * (2.5 - (-2.5)) - 2.5$$

One type of sigma-delta ADC

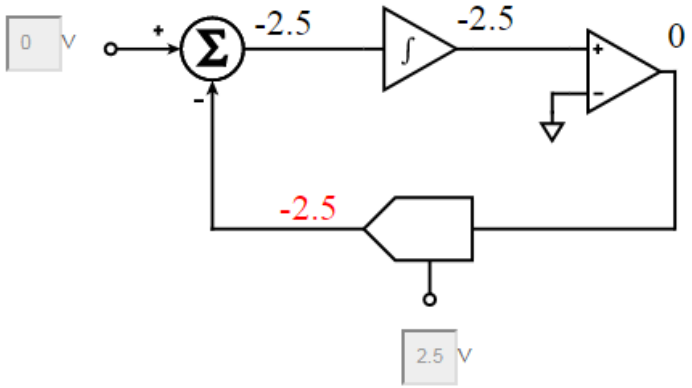


Bit Stream: 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

Mean Output: 2.5

The comparator output is 1, so the DAC output becomes 2.5

Start Over | Previous Step | Next Step | Next 512 Loops



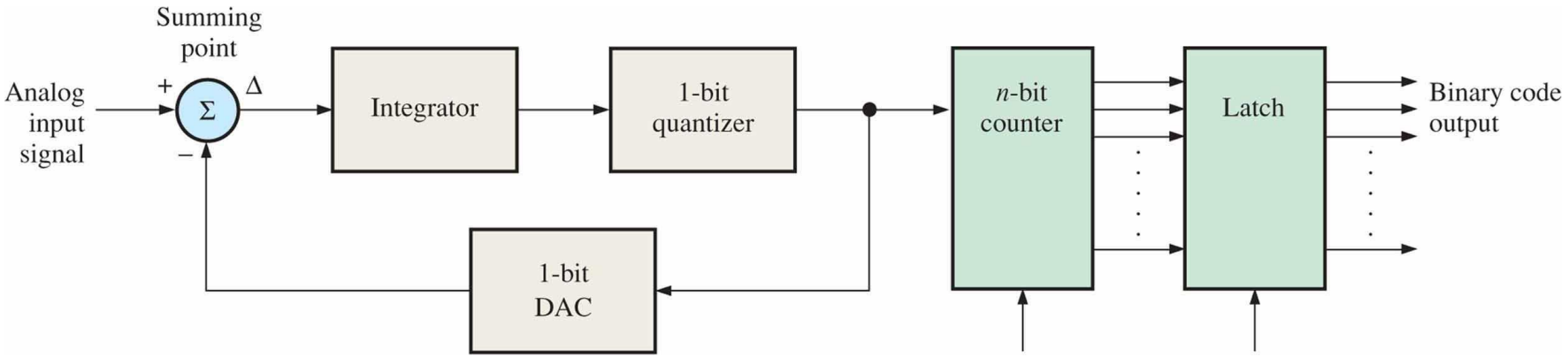
Bit Stream: 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0

Mean Output: 0

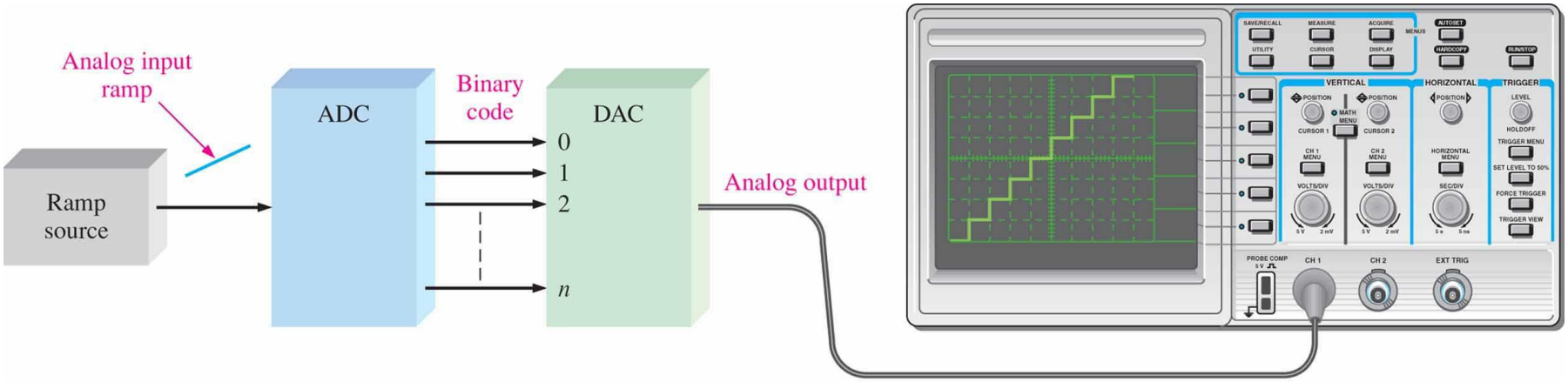
The comparator output is 0, so the DAC output becomes -2.5

Start Over | Previous Step | Next Step | Next 512 Loops

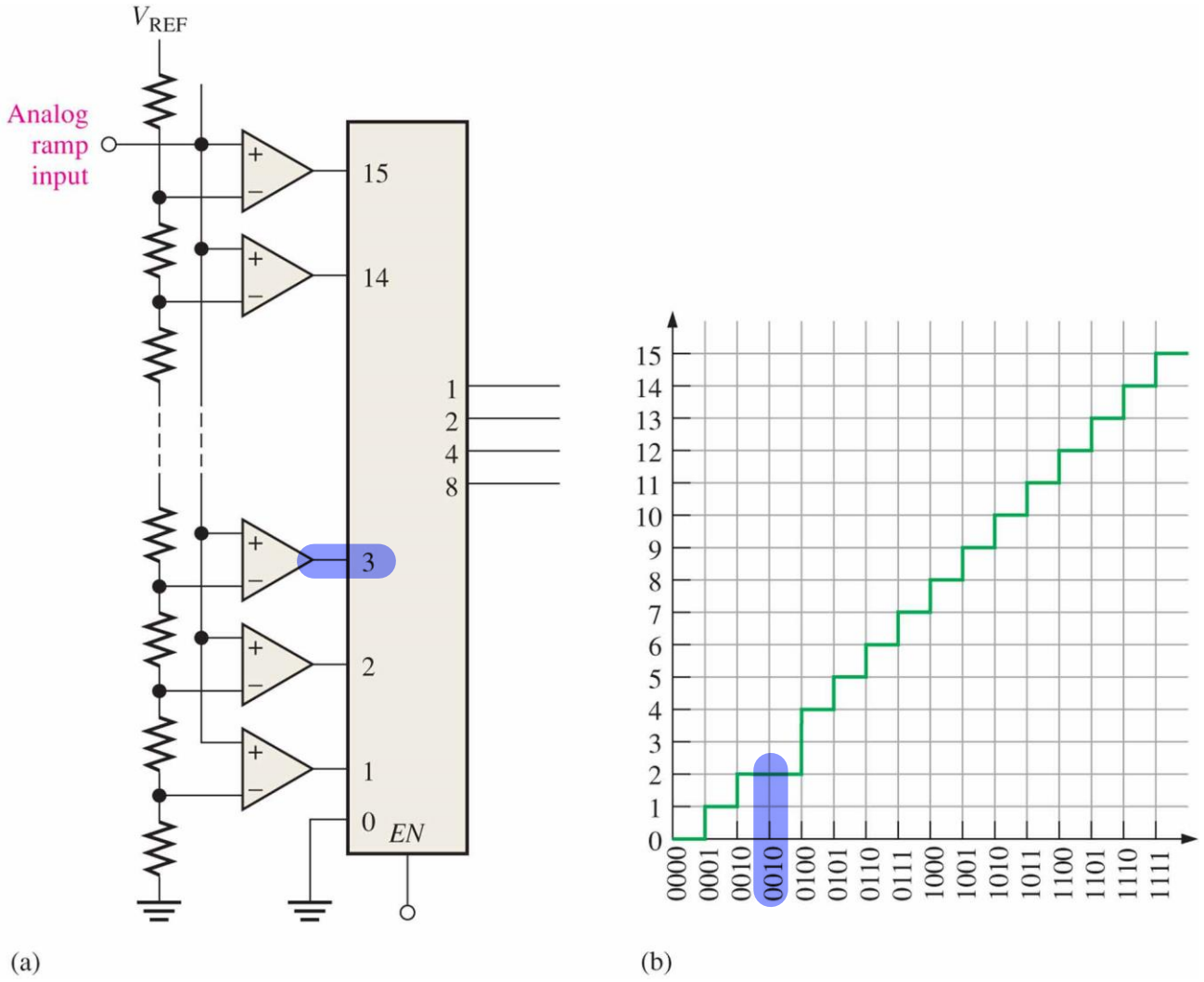
One type of sigma-delta ADC



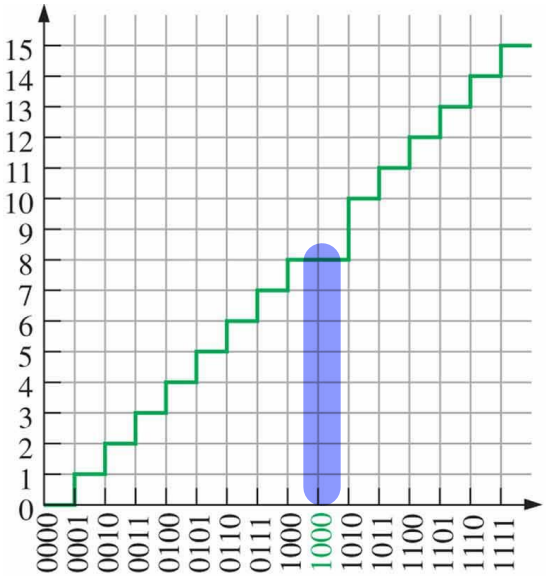
Method for testing ADCs



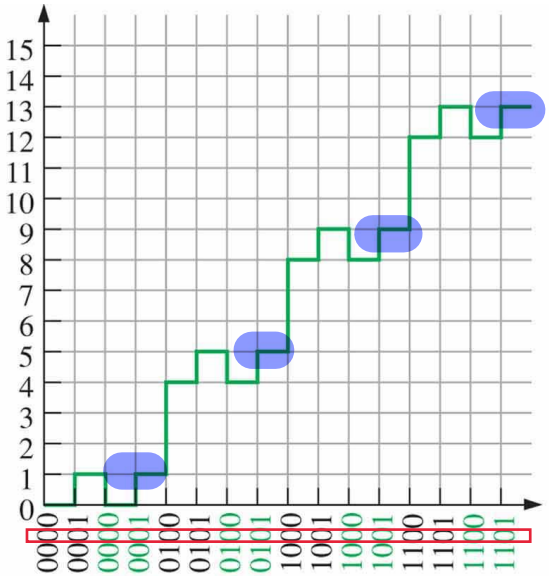
ADC and its output



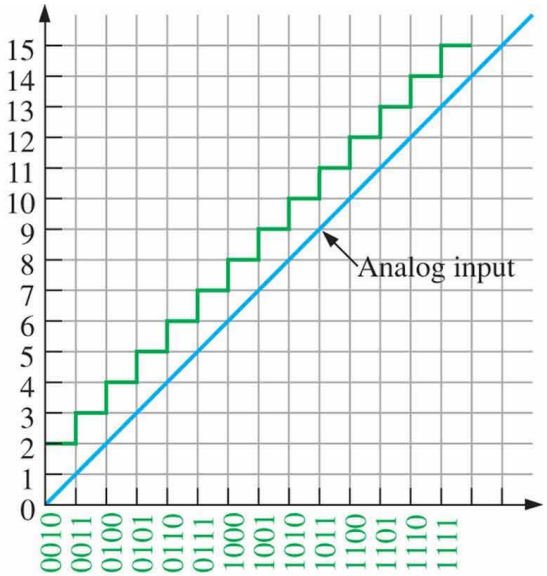
Analog-to-digital conversion errors



(a) Missing code (green)



(b) Incorrect codes (green)

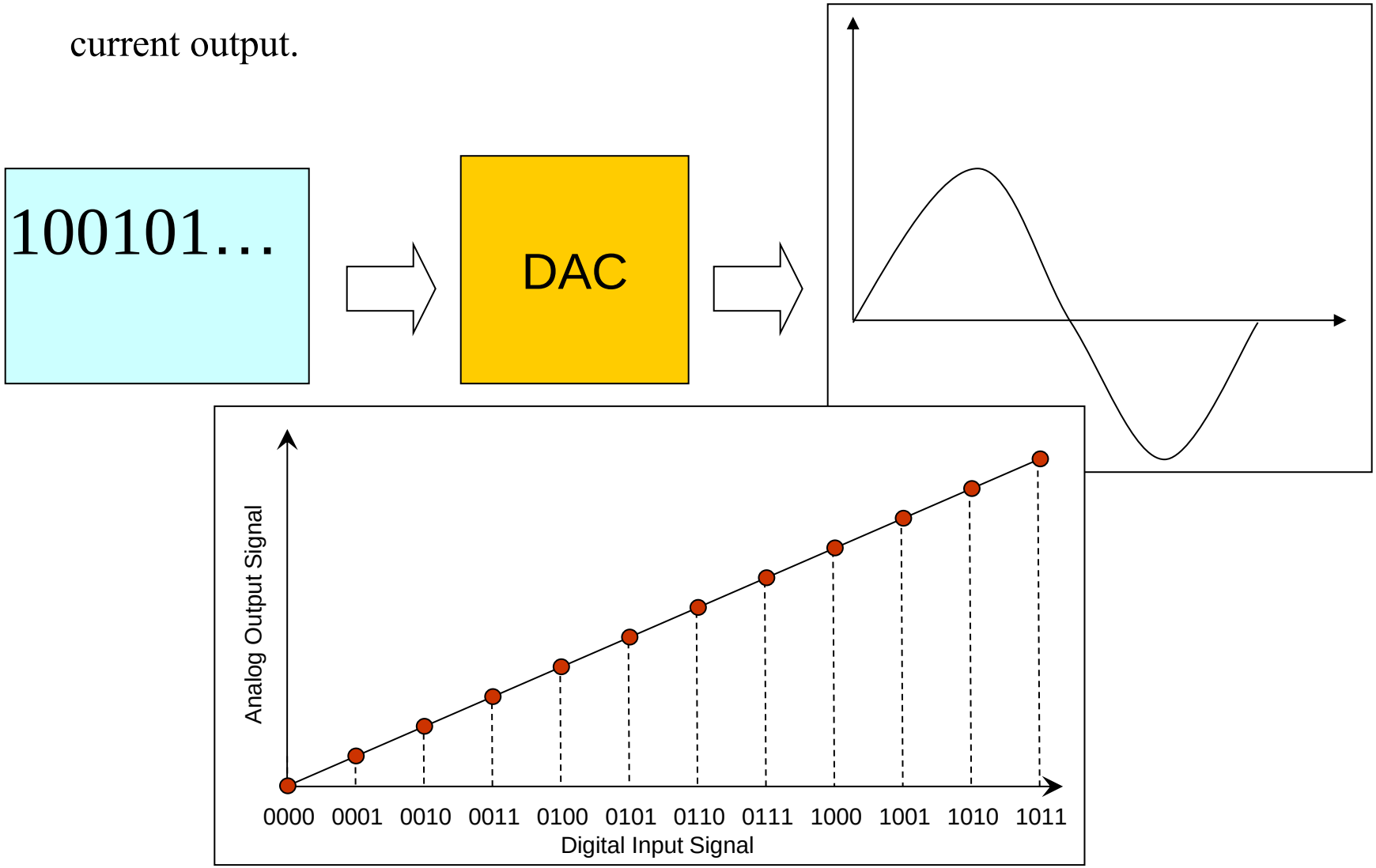


(c) Offset

Digital-to-Analog Converter

DAC

A digital to analog converter (DAC) converts a digital signal to an analog voltage or current output.



Types of DACs

Many types of DACs available.

Usually switches, resistors, and op-amps used to implement conversion

Two Types:

- Binary Weighted Resistor
- R-2R Ladder

Binary Weighted Resistor

Utilizes a summing op-amp circuit

Weighted resistors are used to distinguish each bit from the most significant to the least significant

Transistors are used to switch between V_{ref} and ground (bit high or low)

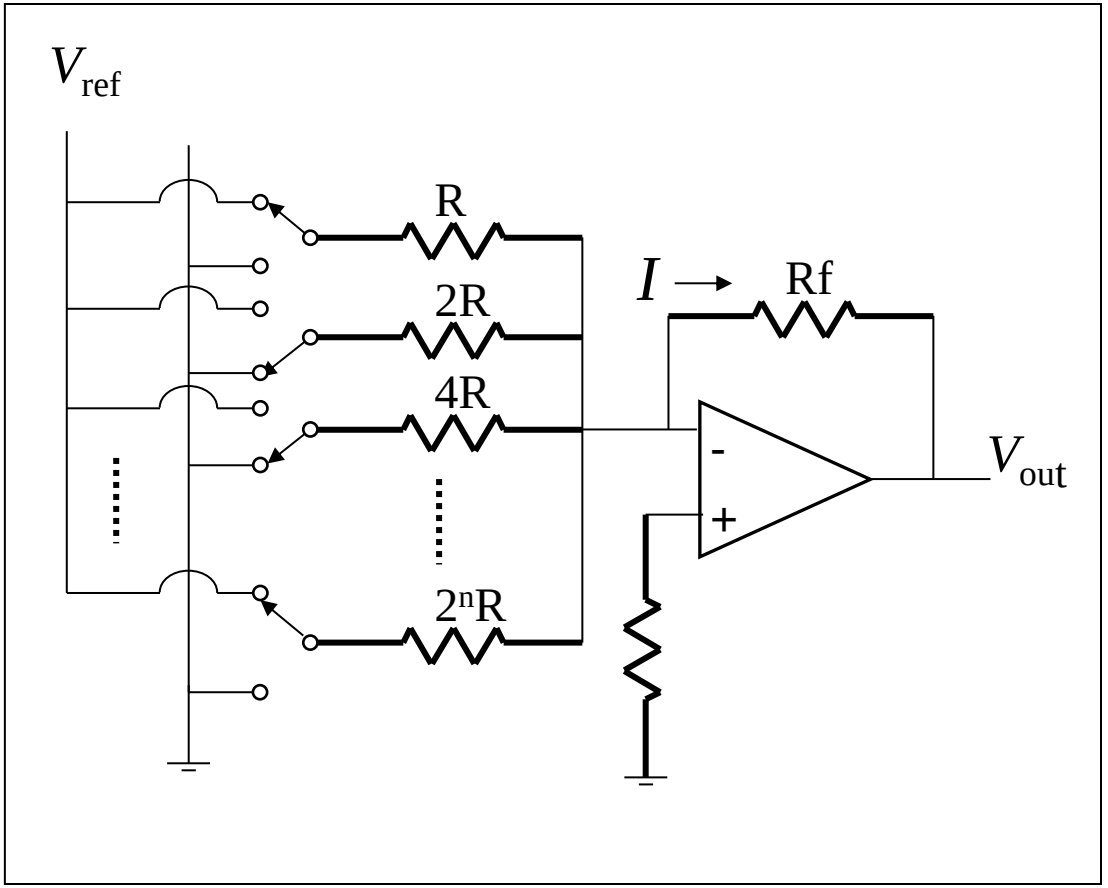
Binary Weighted Resistor

Assume Ideal Op-amp

No current into op-amp

Virtual ground at inverting input

$V_{out} = -IR_f$

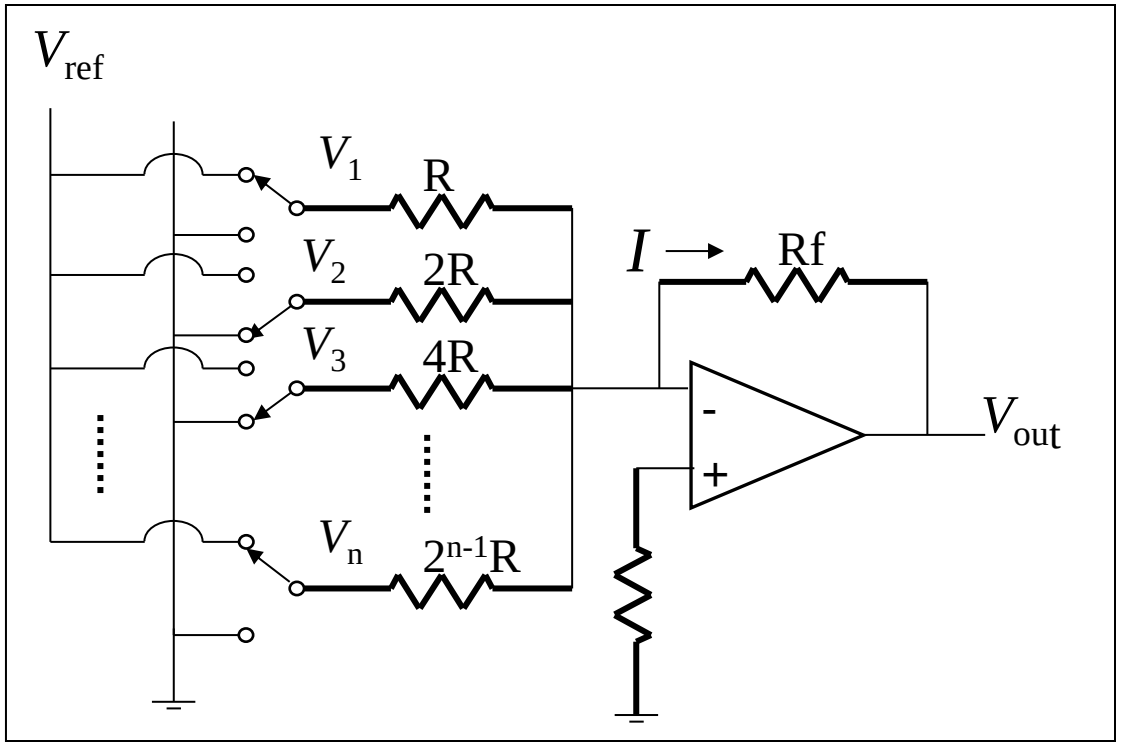


Binary Weighted Resistor

Voltages V_1 through V_n are either V_{ref} if corresponding bit is high or ground if corresponding bit is low

V_1 is most significant bit

V_n is least significant bit



MSB

LSB

$$V_{out} = -IR_f = -R_f\left(\frac{V_1}{R} + \frac{V_2}{2R} + \frac{V_3}{4R} + \cdots \frac{V_n}{2^{n-1}R}\right)$$

Binary Weighted Resistor

If $R_f = R/2$

$$V_{out} = -IR_f = -\left(\frac{V_1}{2} + \frac{V_2}{4} + \frac{V_3}{8} + \cdots \frac{V_n}{2^n}\right)$$

For example, a 4-Bit converter yields

$$V_{out} = -V_{ref}\left(b_3\frac{1}{2} + b_2\frac{1}{4} + b_1\frac{1}{8} + b_0\frac{1}{16}\right)$$

Where b_3 corresponds to Bit-3, b_2 to Bit-2, etc.

Binary Weighted Resistor

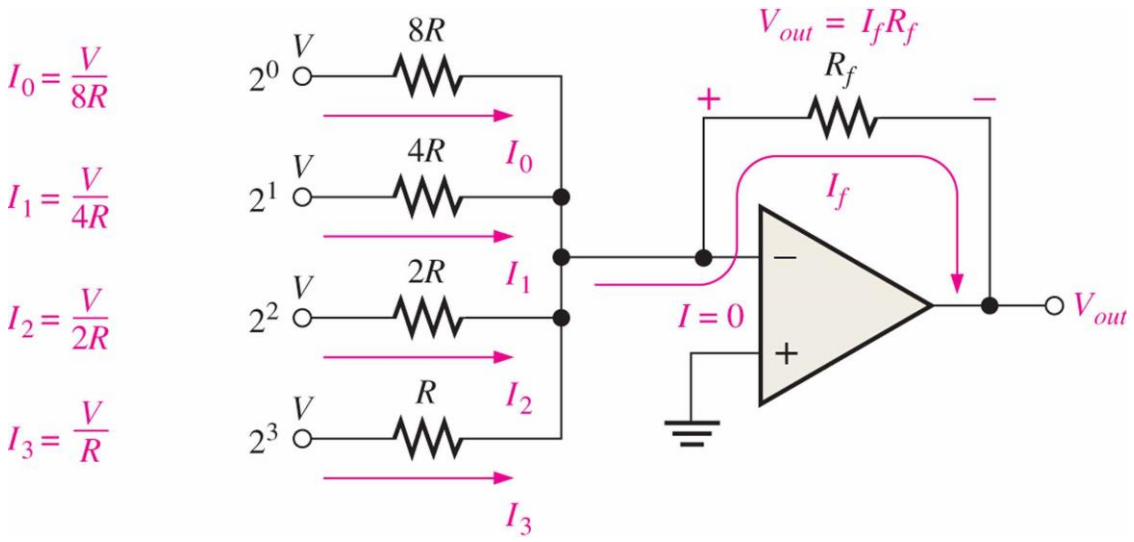
Advantages

- Simple Construction/Analysis
- Fast Conversion

Disadvantages

- Requires large range of resistors (2000:1 for 12-bit DAC) with necessary high precision for low resistors
- Requires low switch resistances in transistors
- Can be expensive. Therefore, usually limited to 8-bit resolution.

4-bit DAC with binary-weighted inputs



$$V_{out} = I_f R_f$$

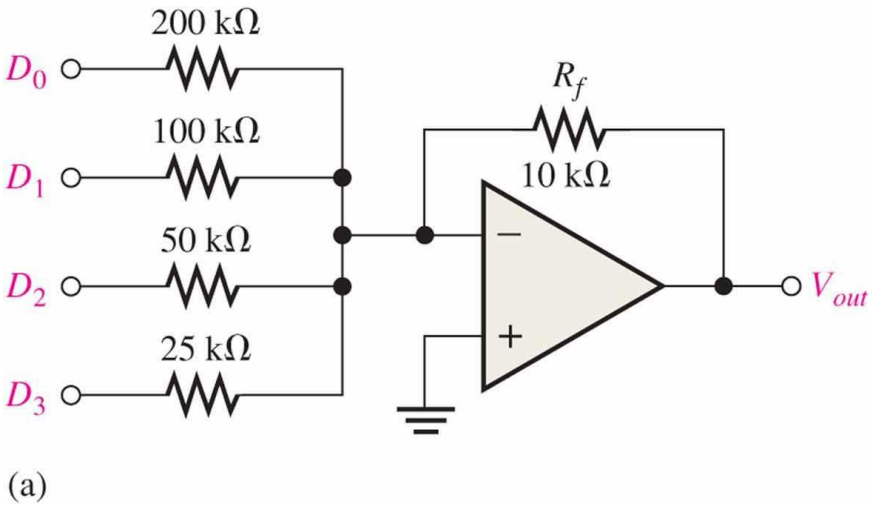
$$= \left(\sum_{i=0}^{N-1} I_i \right) R_f = -\left(\frac{V_0}{8R} + \frac{V_1}{4R} + \frac{V_2}{2R} + \frac{V_3}{1R} \right) R_f$$

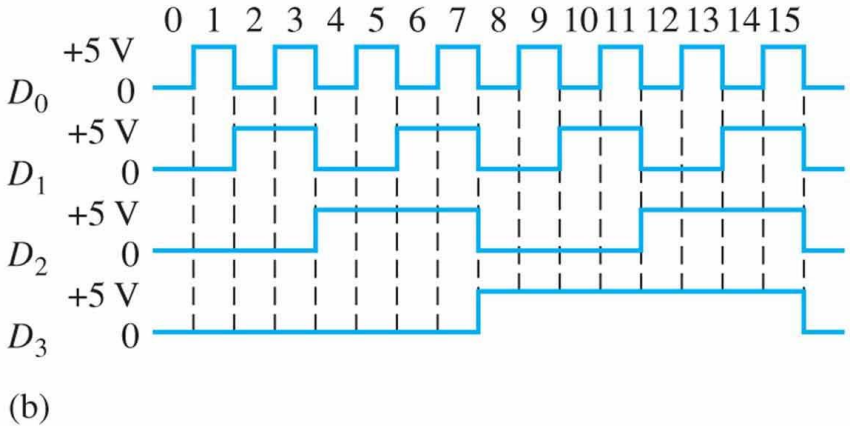
$$= -R_f \left(\frac{V_0}{2^{N-1}R} + \frac{V_1}{2^{N-2}R} + \cdots + \frac{V_{N-2}}{2R} + \frac{V_{N-1}}{R} \right)$$

$$= -\frac{R_f}{R} \left(V_0 2^{-(N-1)} + V_1 2^{-(N-2)} + \cdots + V_{N-2} 2^{-1} + V_{N-1} 2^{-0} \right)$$

$$= -\frac{R_f 2^{-(N-1)}}{R} \left(V_0 2^0 + V_1 2^1 + \cdots + V_{N-2} 2^{N-2} + V_{N-1} 2^{N-1} \right)$$

4-bit DAC with binary-weighted inputs





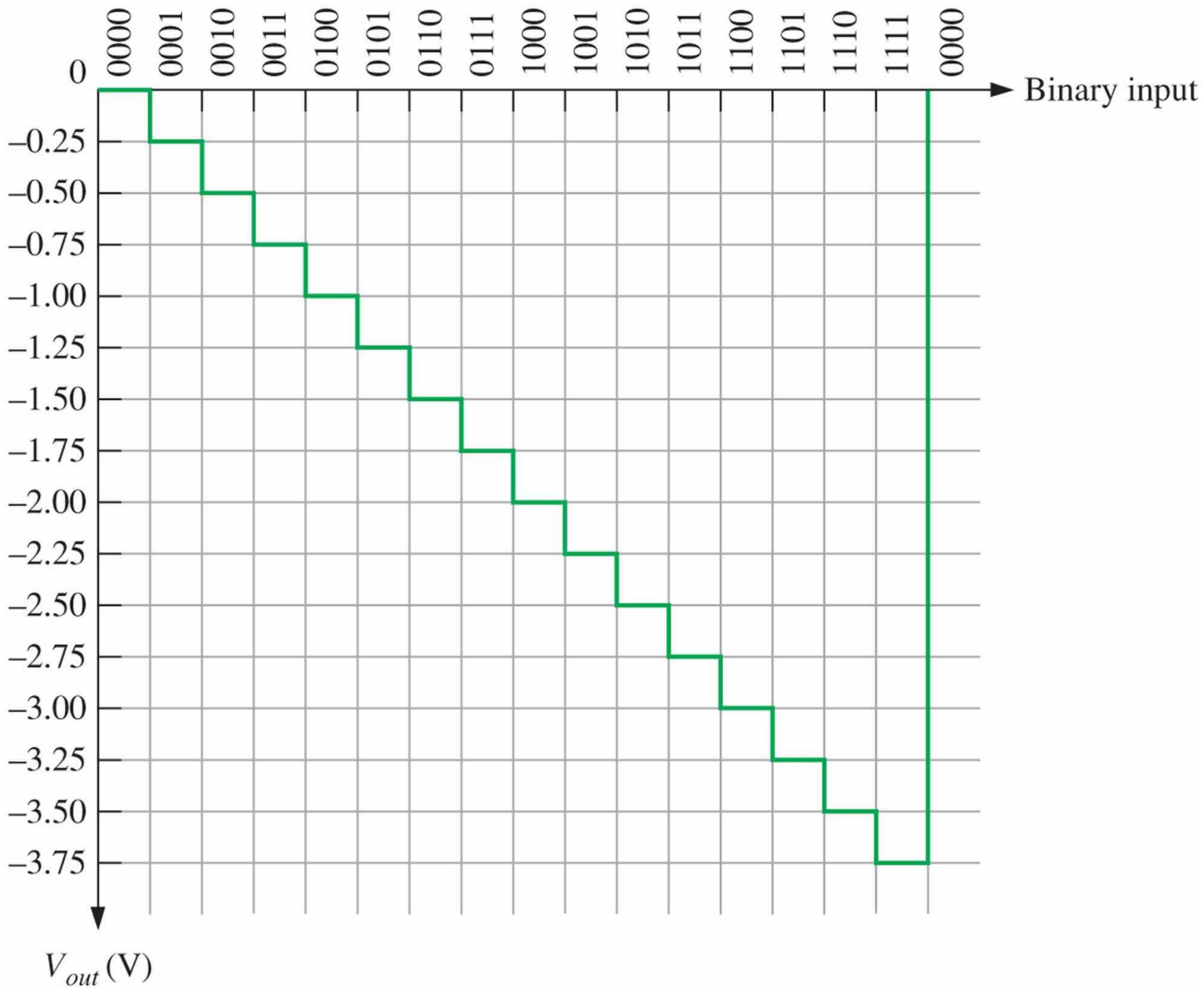
$$V_{out} = -\frac{1}{2^{(N-1)}}\left(\frac{R_f}{R}\right)\left(V_0 2^0 + V_1 2^1 + \cdots + V_{N-2} 2^{N-2} + V_{N-1} 2^{N-1}\right);$$

$$when; \quad N = 4; R_f = 10k\Omega; R = 25k\Omega; V_i = D_i * 5V; D_i \in \{0,1\}$$

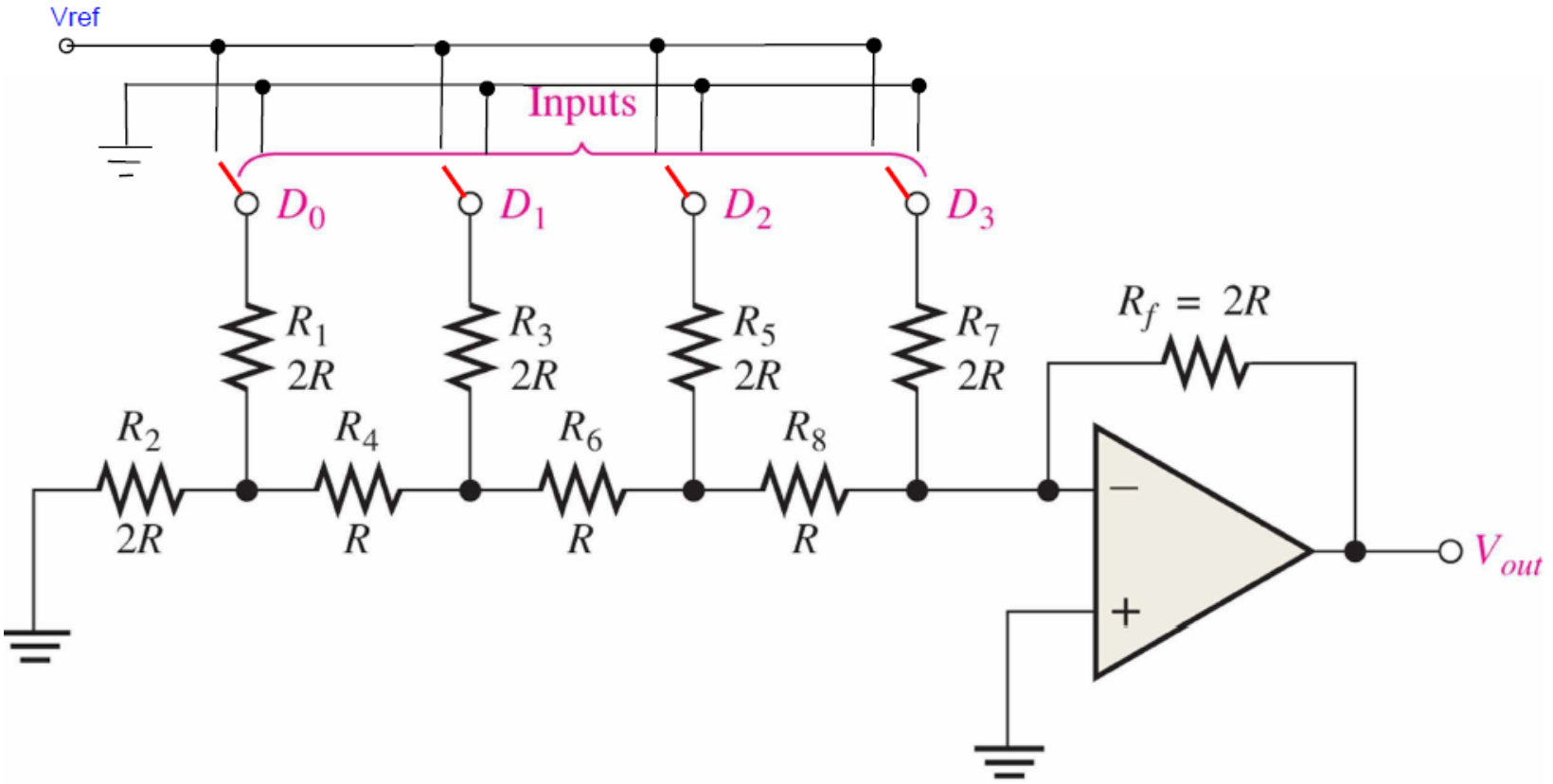
$$V_{out} = -\frac{1}{8}\left(\frac{10}{25}\right)V\left(D_0 2^0 + D_1 2^1 + D_2 2^2 + D_3 2^3\right); V = 5$$

$$= -0.25 * \left(D_0 2^0 + D_1 2^1 + D_2 2^2 + D_3 2^3\right)$$

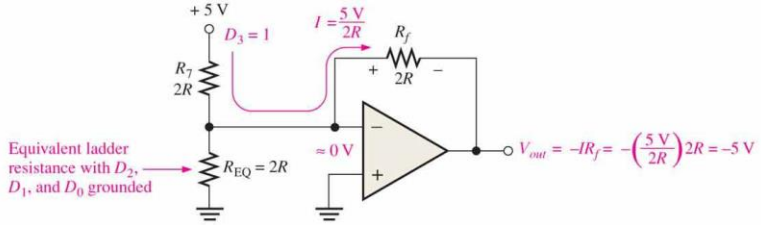
4-bit DAC with binary-weighted inputs; Output of the DAC



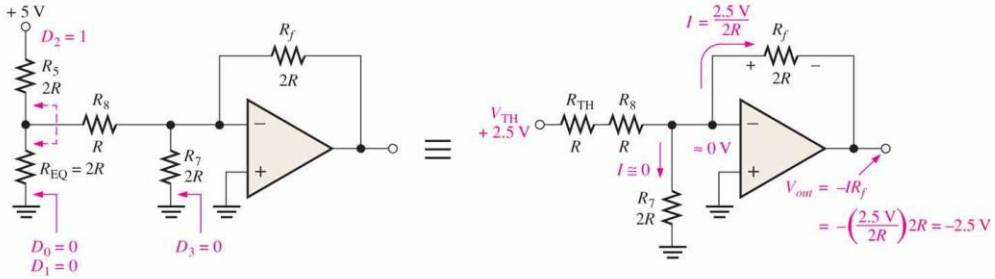
An R/2R ladder DAC



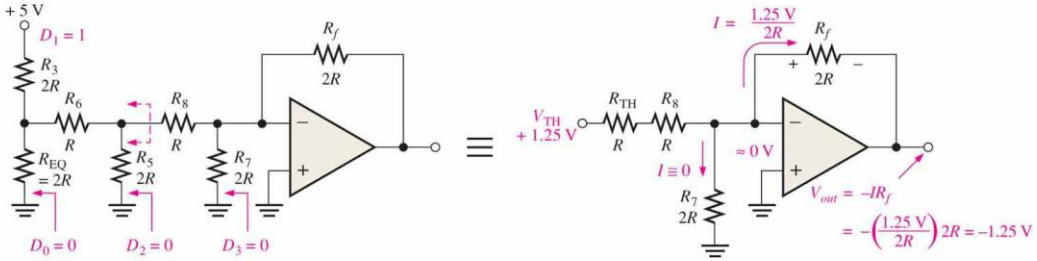
Analysis of the R/2R ladder DAC



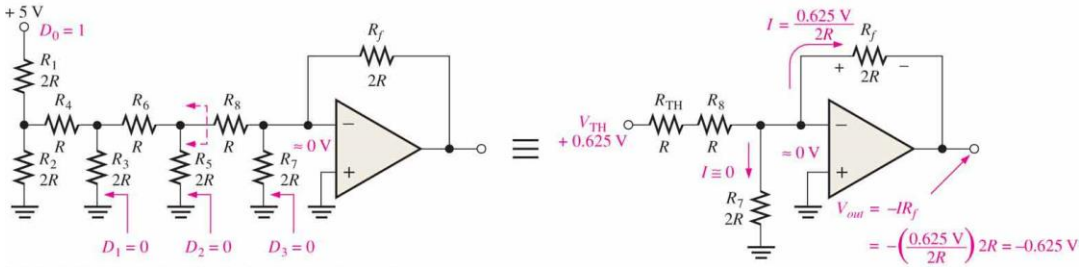
(a) Equivalent circuit for $D_3 = 1, D_2 = 0, D_1 = 0, D_0 = 0$



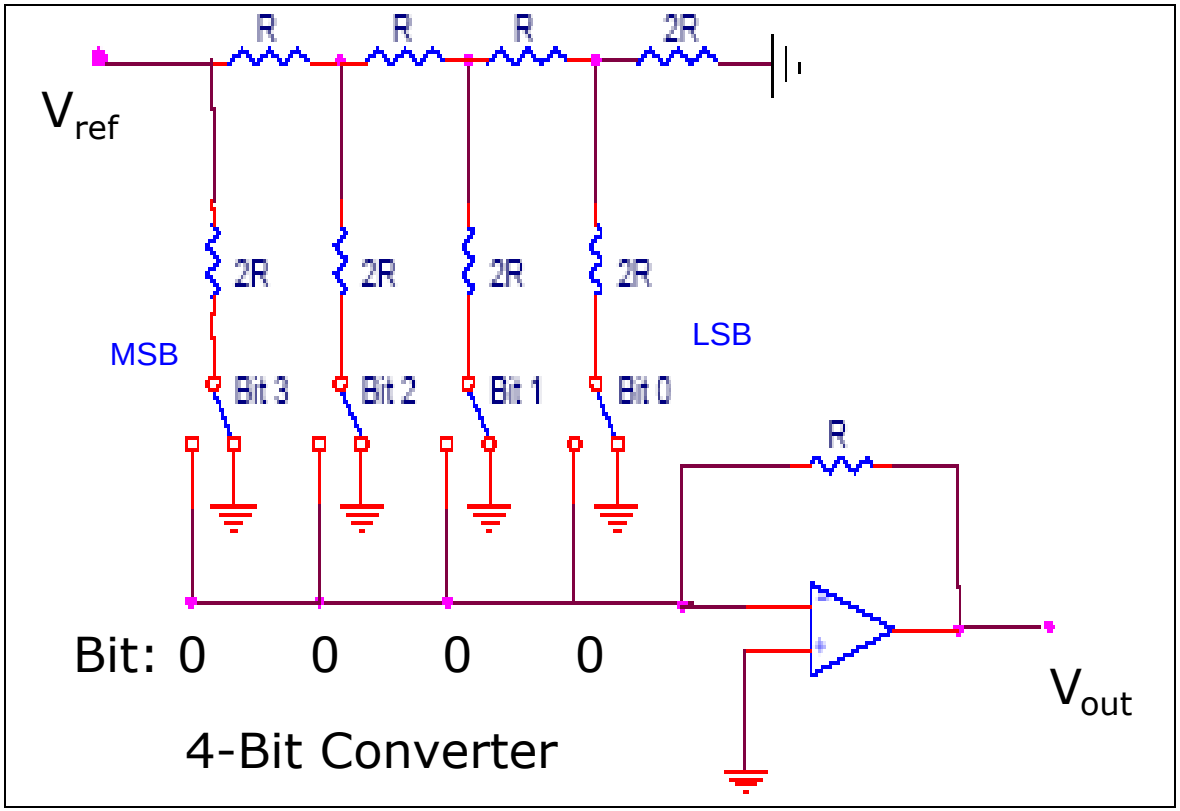
(b) Equivalent circuit for $D_3 = 0, D_2 = 1, D_1 = 0, D_0 = 0$



(c) Equivalent circuit for $D_3 = 0, D_2 = 0, D_1 = 1, D_0 = 0$



R-2R Ladder

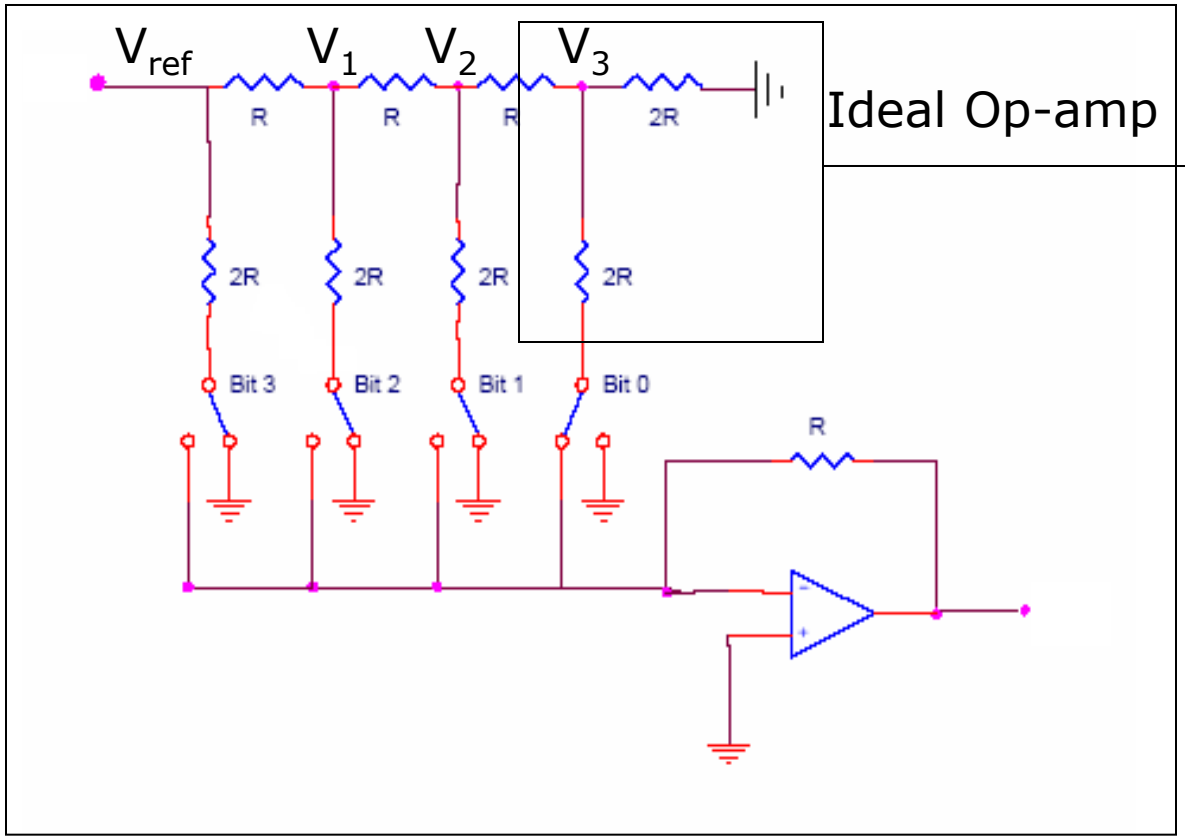


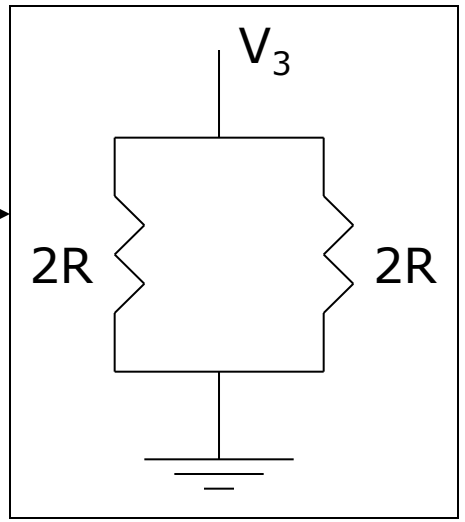
Each bit corresponds to a switch:

If the bit is high, the corresponding switch is connected to the inverting input of the op-amp.

If the bit is low, the corresponding switch is connected to ground.

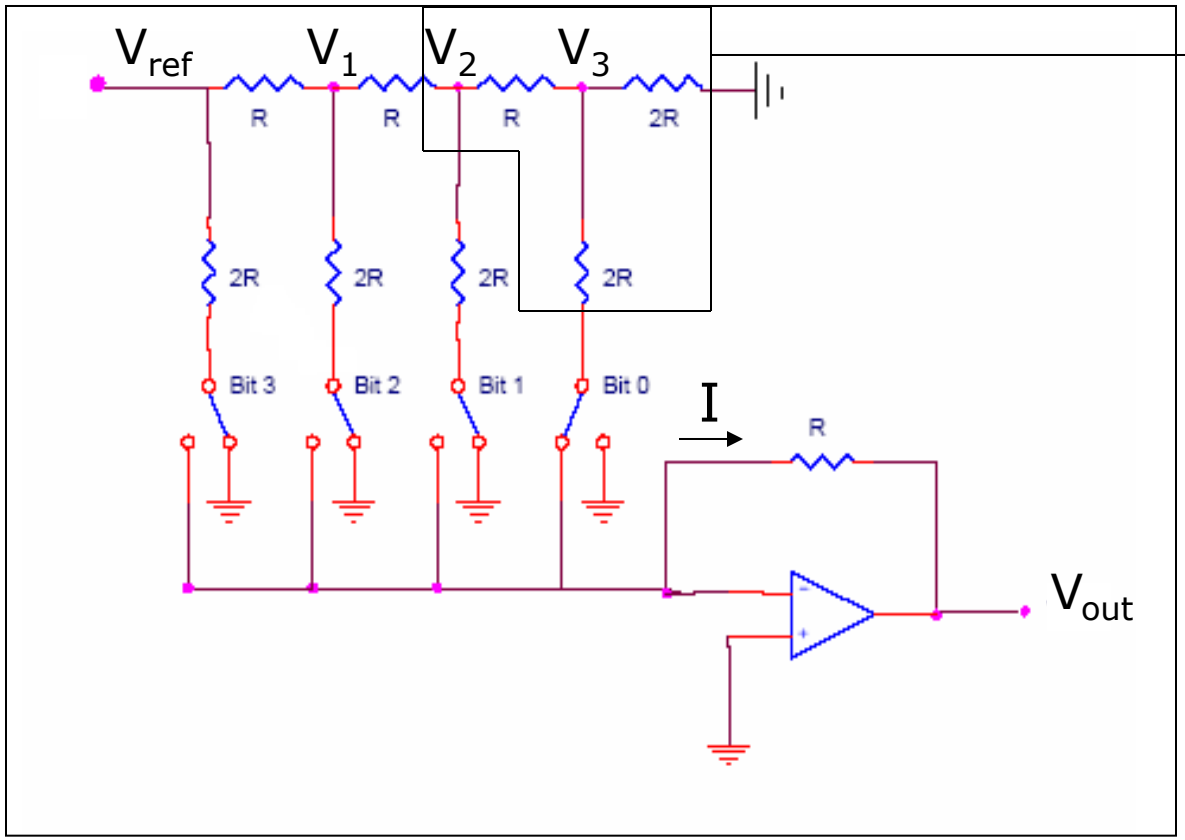
R-2R Ladder

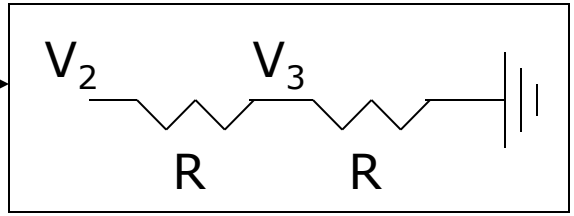




$$R_{eq} = \frac{(2R)(2R)}{(2R+2R)} = R$$

R-2R Ladder





$$V_3 = \left(\frac{R}{R+R}\right)V_2 = \frac{1}{2}V_2$$

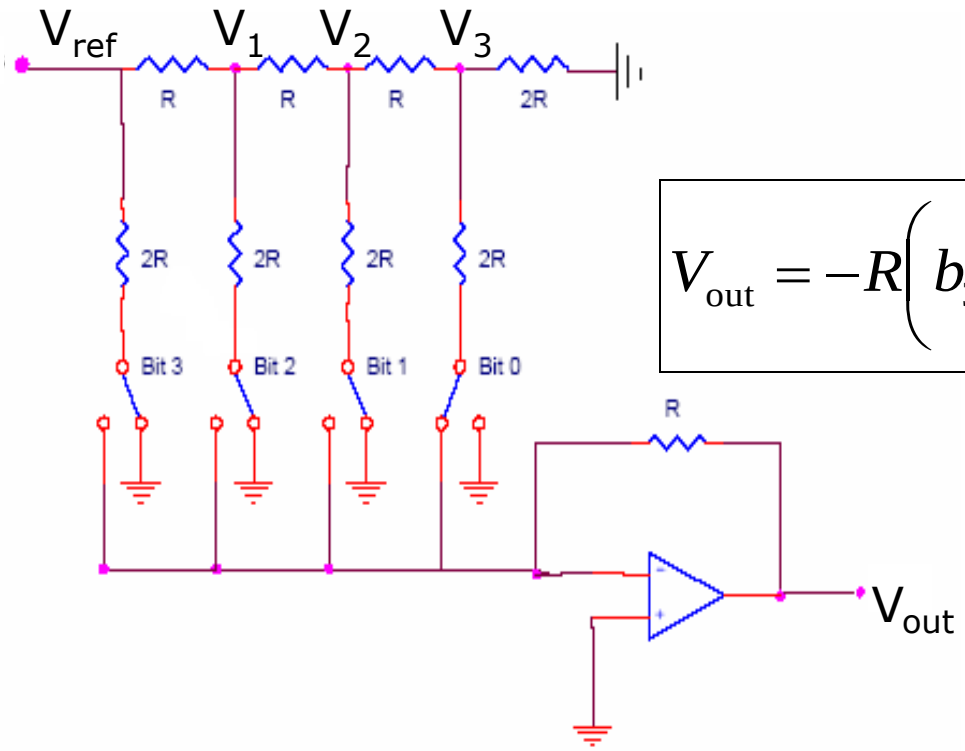
Likewise,

$$V_2 = \frac{1}{2}V_1$$

$$V_1 = \frac{1}{2}V_{\text{ref}}$$

$$V_{\text{out}} = -IR$$

R-2R Ladder



Results:

$$V_3 = \frac{1}{8}V_{\text{ref}},\ V_2 = \frac{1}{4}V_{\text{ref}},\ V_1 = \frac{1}{2}V_{\text{ref}}$$

$$V_{\text{out}} = -R\left(b_3 \frac{V_{\text{ref}}}{2R} + b_2 \frac{V_{\text{ref}}}{4R} + b_1 \frac{V_{\text{ref}}}{8R} + b_0 \frac{V_{\text{ref}}}{16R} \right)$$

Where b_3 corresponds to bit 3, b_2 to bit 2, etc.

If bit n is set, b_n=1

If bit n is clear, b_n=0

R-2R Ladder

For a 4-Bit R-2R Ladder

$$V_{\text{out}} = -V_{\text{ref}}\left(b_3\frac{1}{2}+b_2\frac{1}{4}+b_1\frac{1}{8}+b_0\frac{1}{16}\right)$$

For general n-Bit R-2R Ladder or Binary Weighted Resister DAC

$$V_{\text{out}} = -V_{\text{ref}}\sum_{i=1}^{n} b_{n-i}\frac{1}{2^i}$$

R-2R Ladder

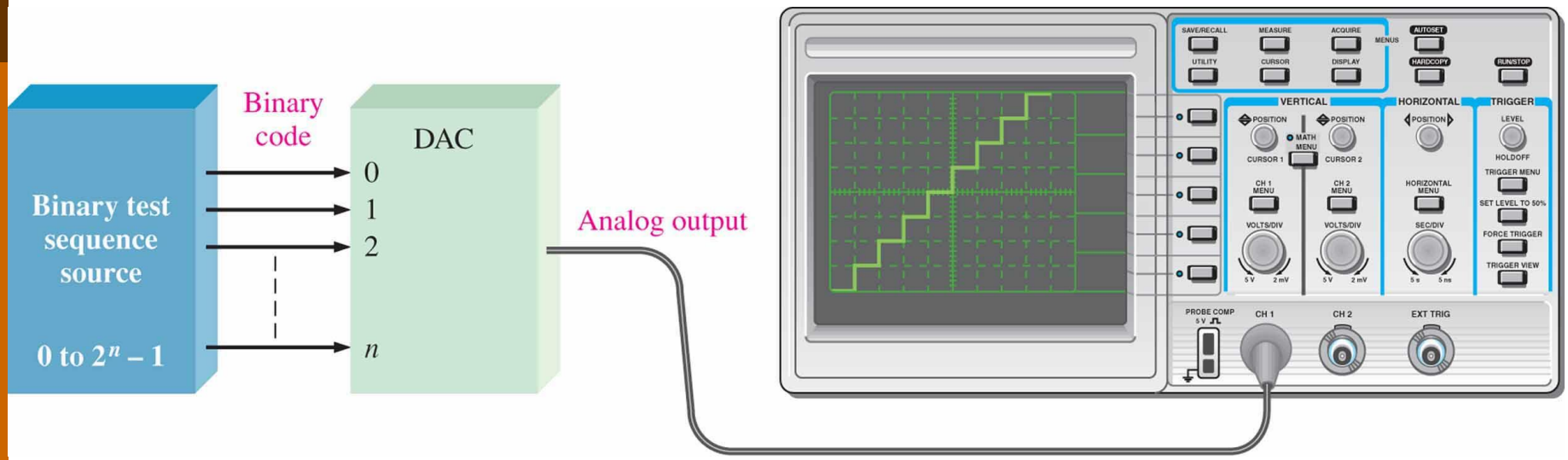
Advantages

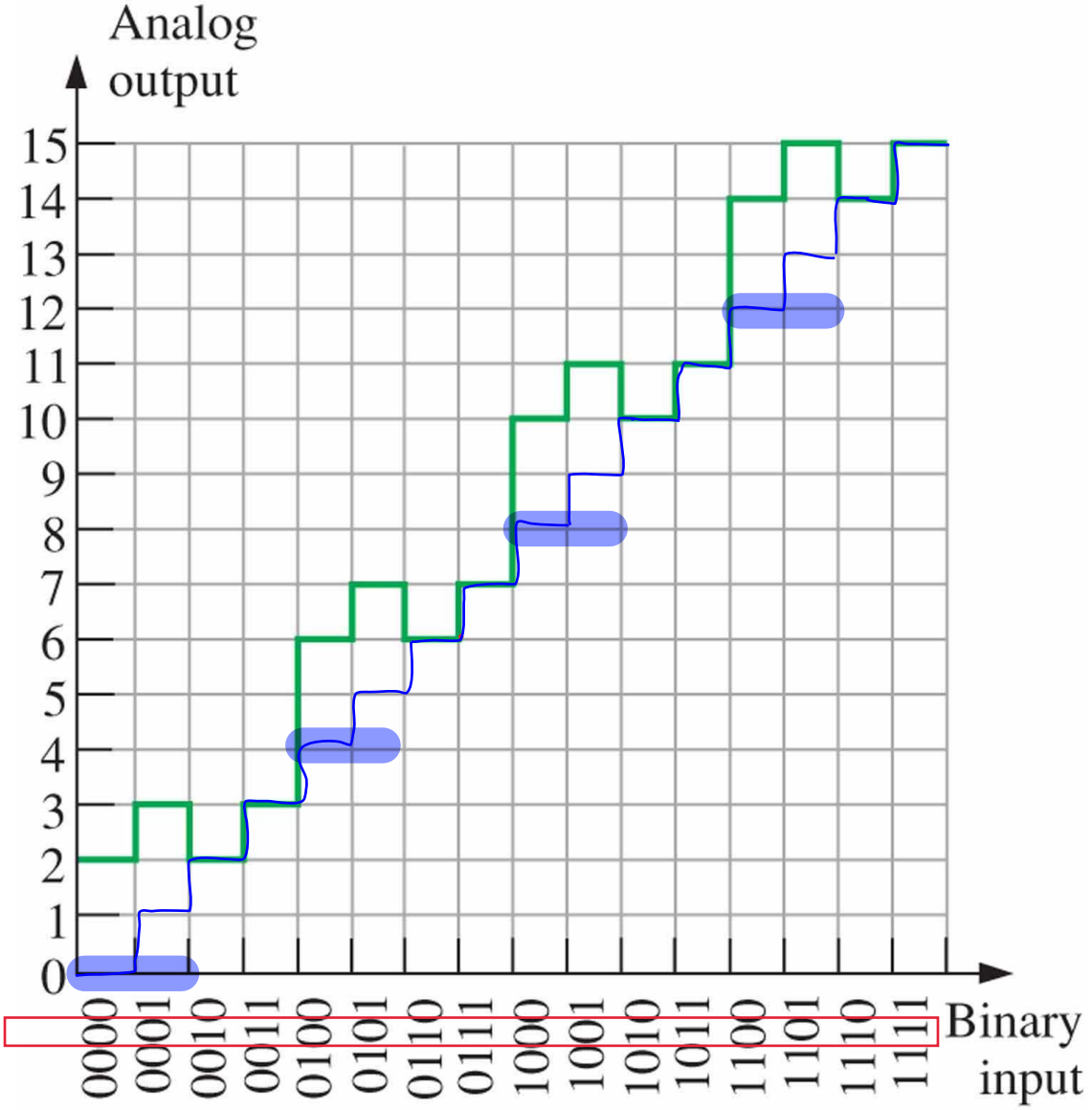
- Only two resistor values (R and 2R)
- Does not require high precision resistors

Disadvantage

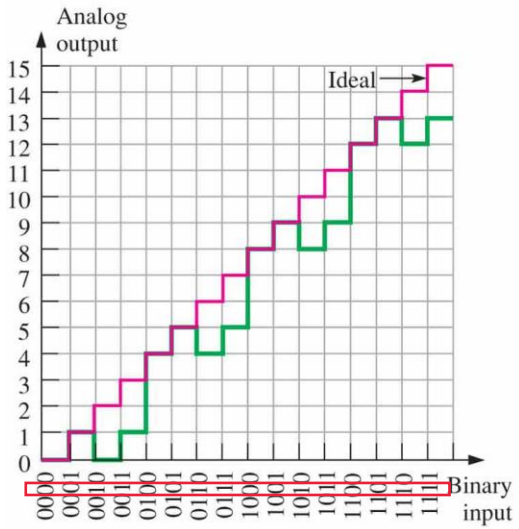
- Lower conversion speed than binary weighted DAC

Basic test setup for a DAC

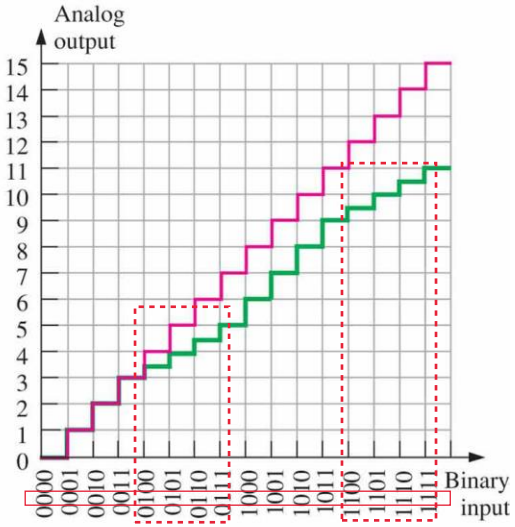



Analog
output
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
0000
0001
0010
0011
0100
0101
0110
0111
1000
1001
1010
1011
1100
1101
1110
1111
Binary
input

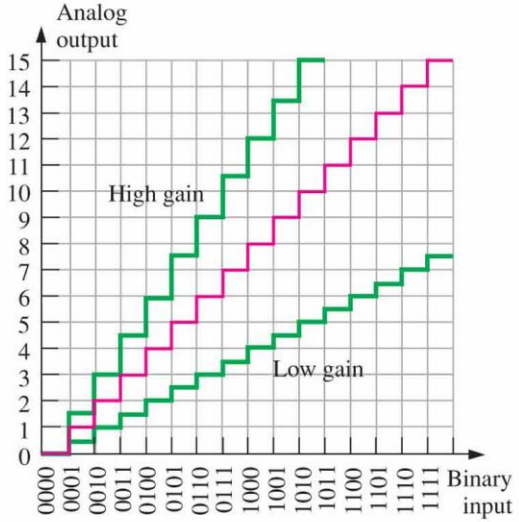
Digital-to-analog conversion errors



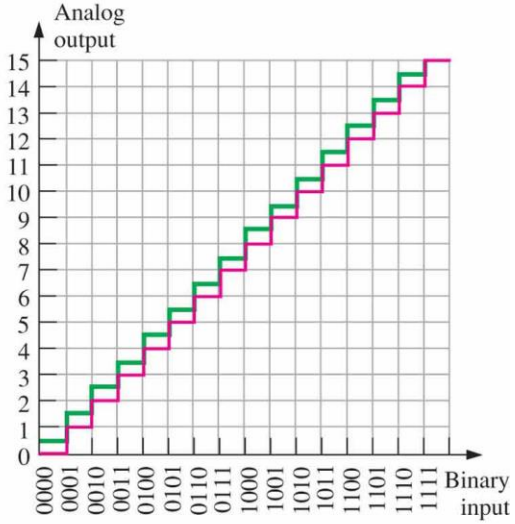
(a) Nonmonotonic output (green)



(b) Differential nonlinearity (green)

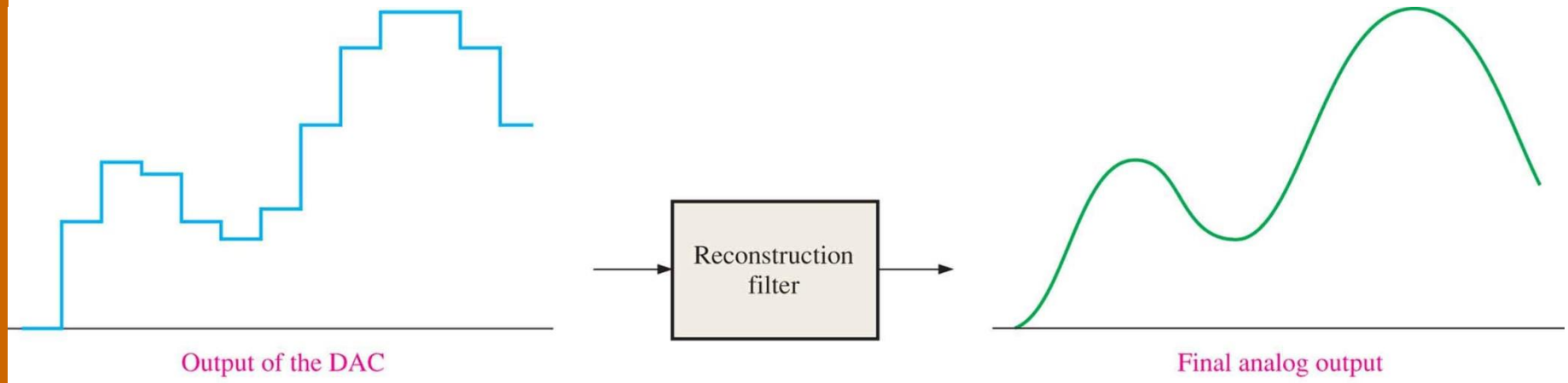


(c) High and low gains (green)

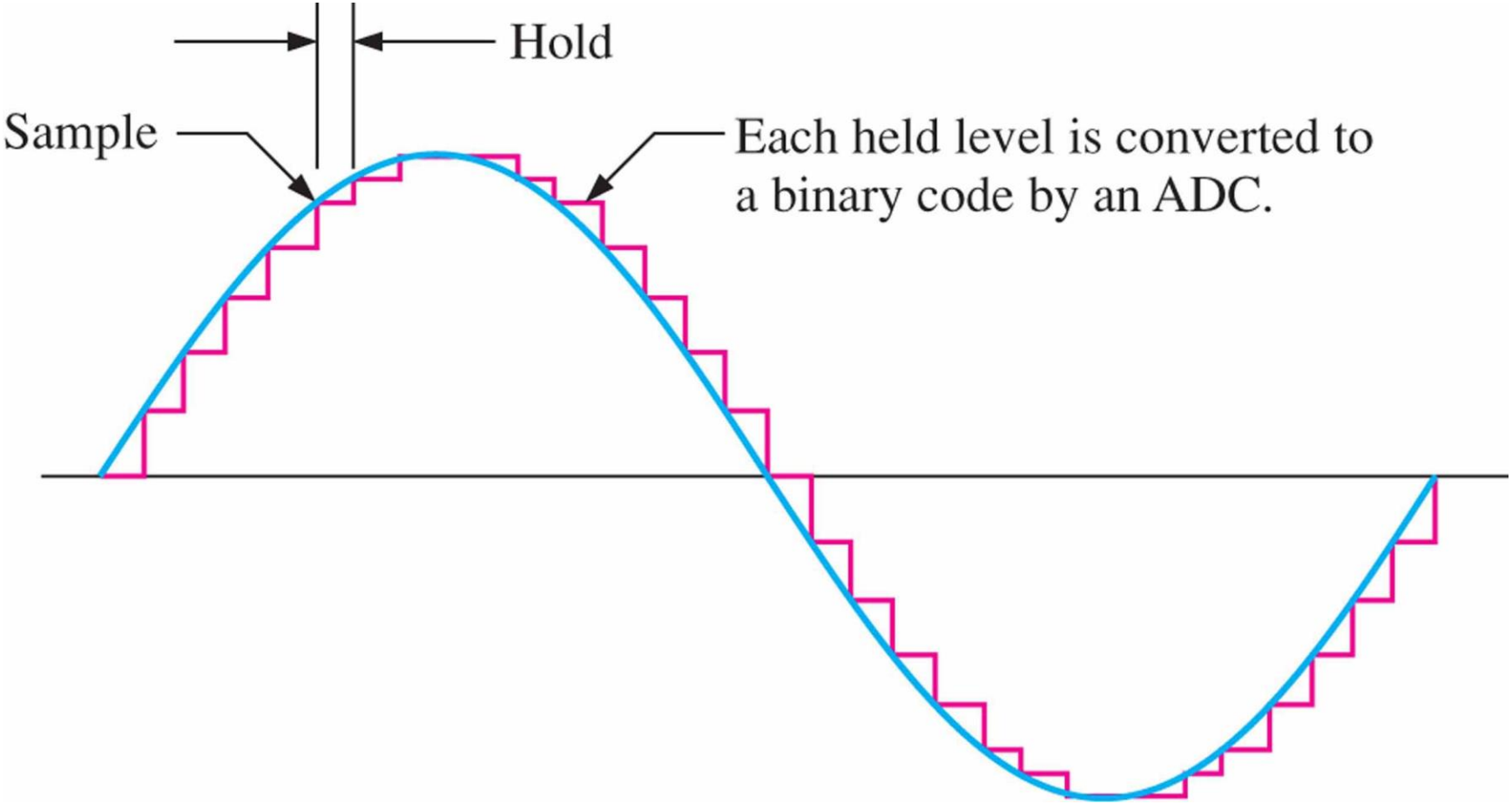


(d) Offset error (green)

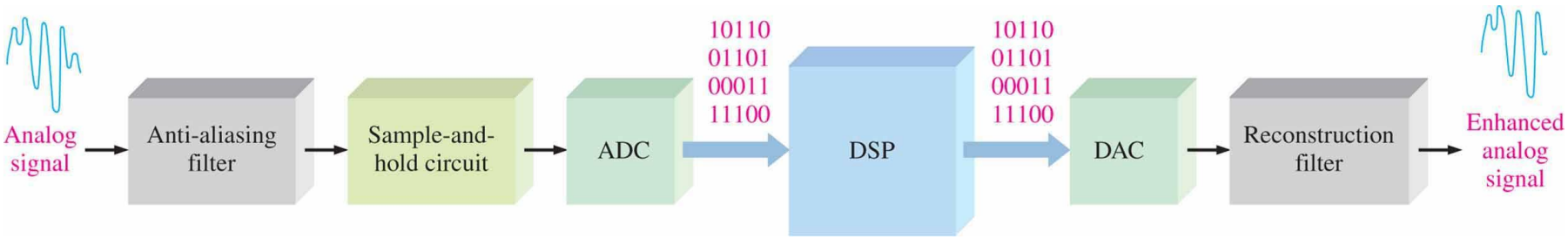
The reconstruction filter smooths the output of the DAC



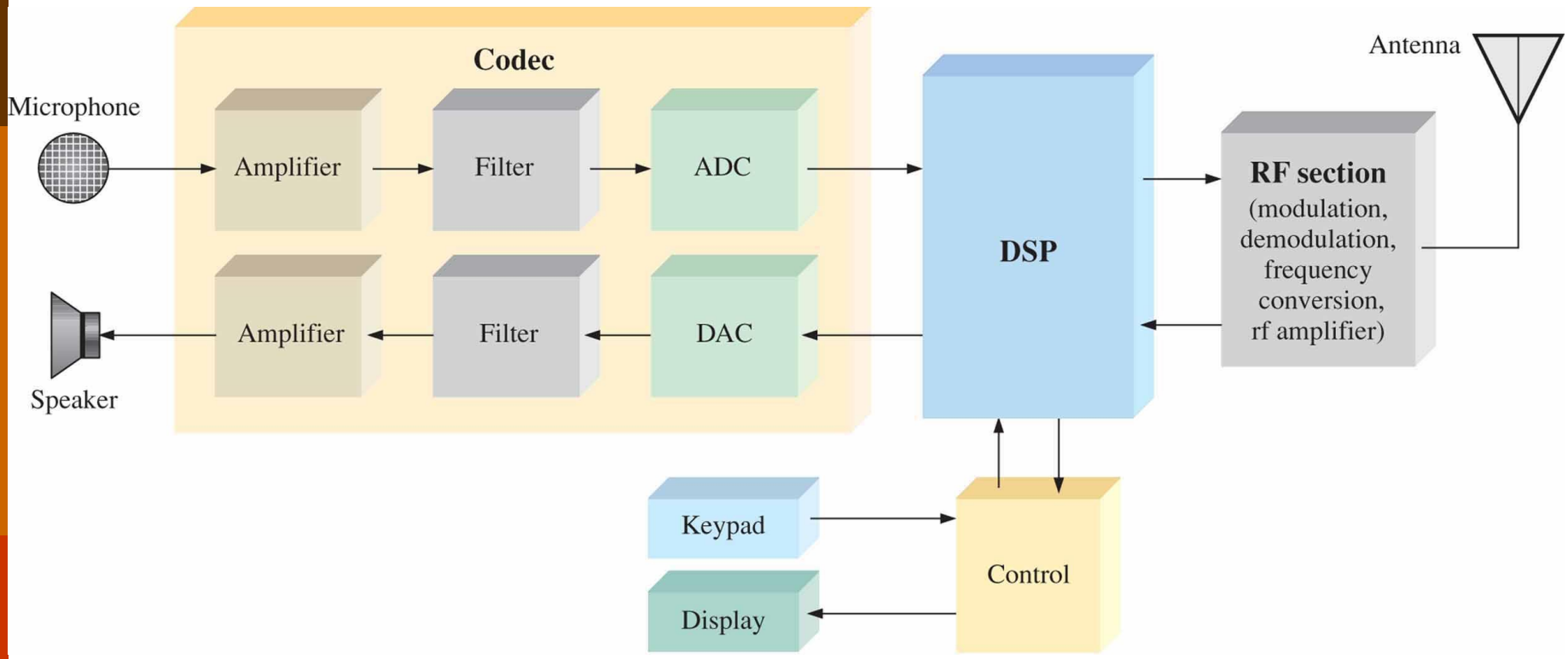
An original analog signal (sine wave) and its "stairstep" approximation



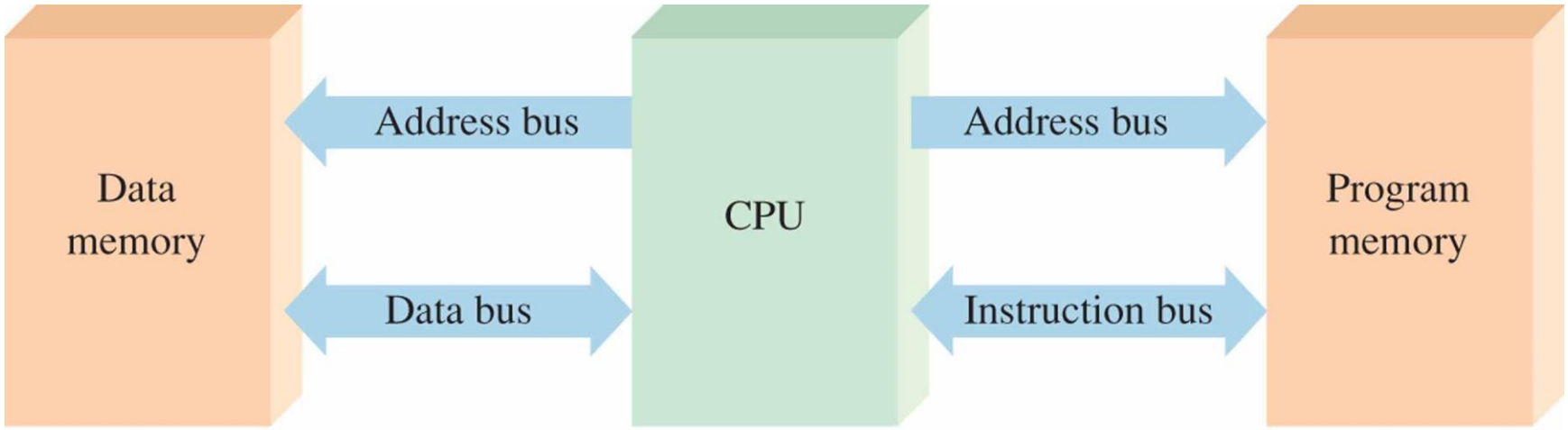
Basic block diagram of a typical digital signal processing system



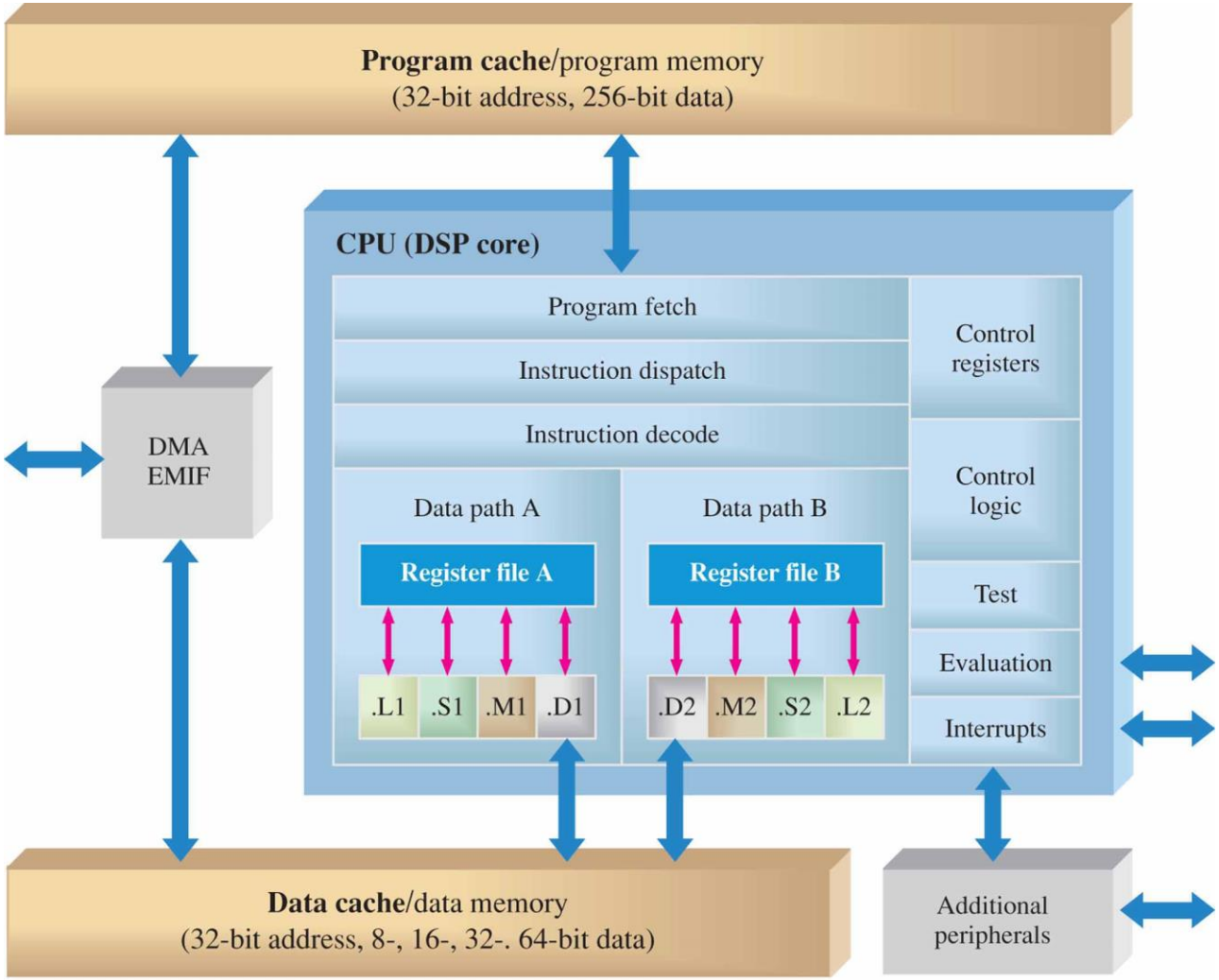
Digital cellular phone



Many DSPs use the Harvard architecture (two memories)



TMS320C6000 series DSP



*DMA: Direct memory access
EMIF: External memory interface

Specifications of DACs

- Resolution
- Speed
- Settling Time
- Reference Voltages
- Linearity
- Errors

Resolution

Smallest analog increment corresponding to 1 LSB change

An N-bit resolution can resolve 2^N distinct analog levels

Common DAC has a 8-16 bit resolution

$$\text{Resolution} = V_{LSB} = \frac{V_{ref}}{2^N}$$

$$\text{where } N = number\ of\ bits$$

Speed

Rate of conversion of a single digital input to its analog equivalent

Conversion rate depends on

- clock speed of input signal
- settling time of converter

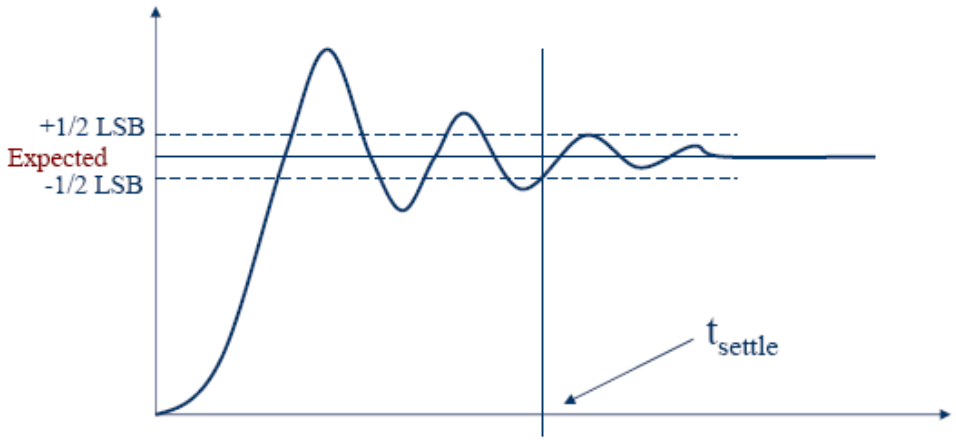
When the input changes rapidly, the DAC conversion speed must be high.

Settling Time

Time required for the output signal to settle within +/- ½ LSB of its final value after a given change in input scale

Limited by slew rate of output amplifier

Ideally, an instantaneous change in analog voltage would occur when a new binary word enters into DAC



Reference Voltages

Used to determine how each digital input will be assigned to each voltage division

Types:

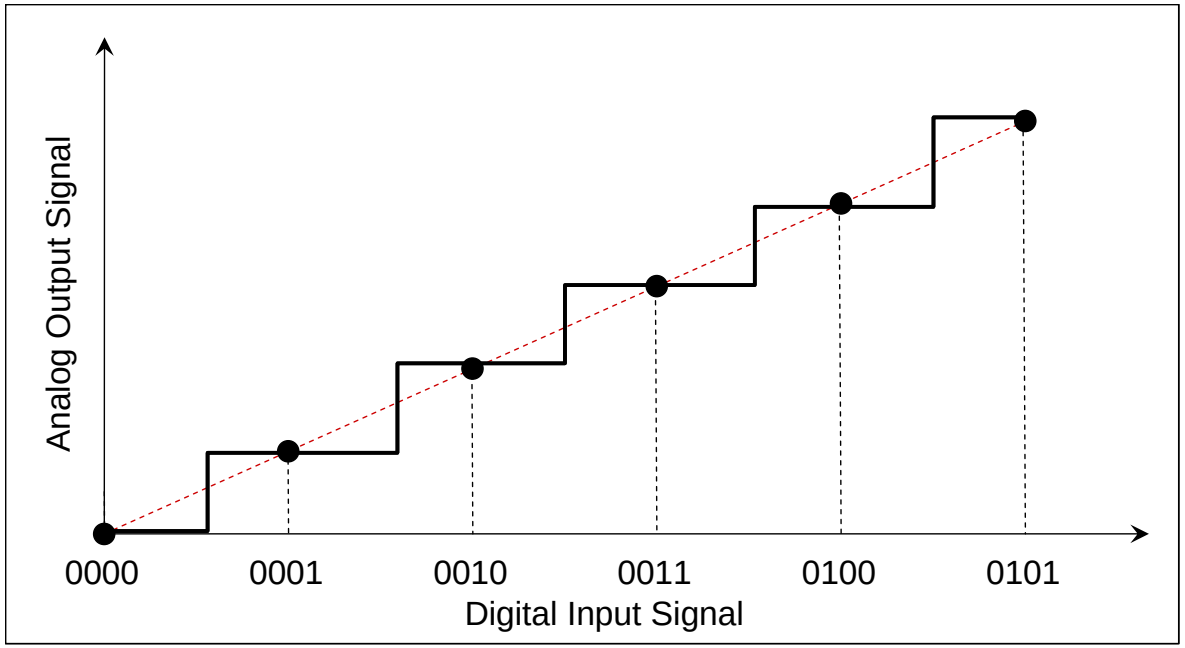
- Non-multiplier DAC: Vref is fixed
- Multiplier DAC: Vref provided by external source

Binary Weighted Resistor

R-2R Ladder

Linearity

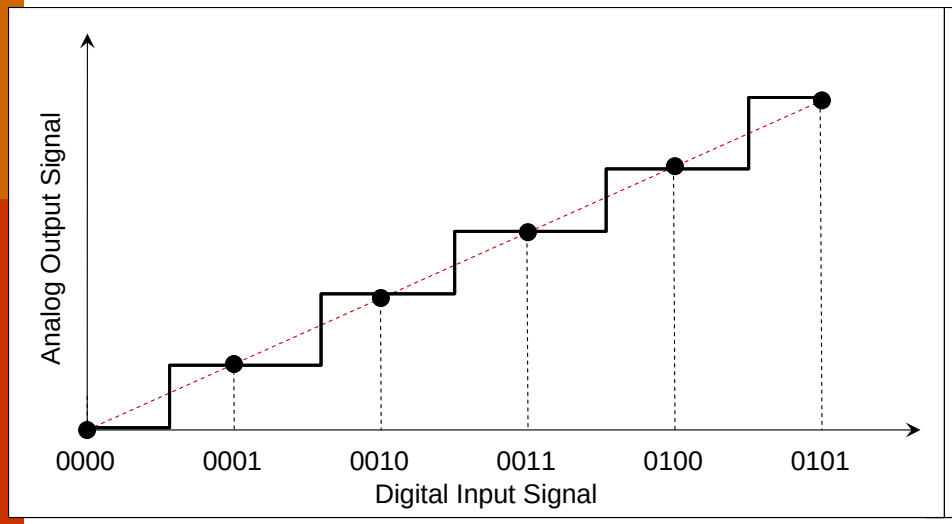
The difference between the desired analog output and the actual output over the full range of expected values



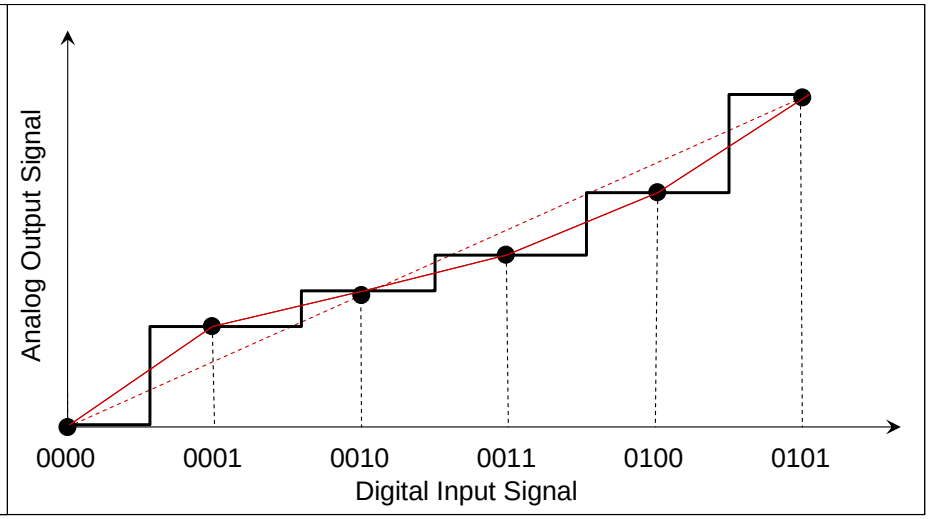
Linearity

Ideally, a DAC should produce a linear relationship between the digital input and analog output

Linearity (Ideal)



Non-Linearity

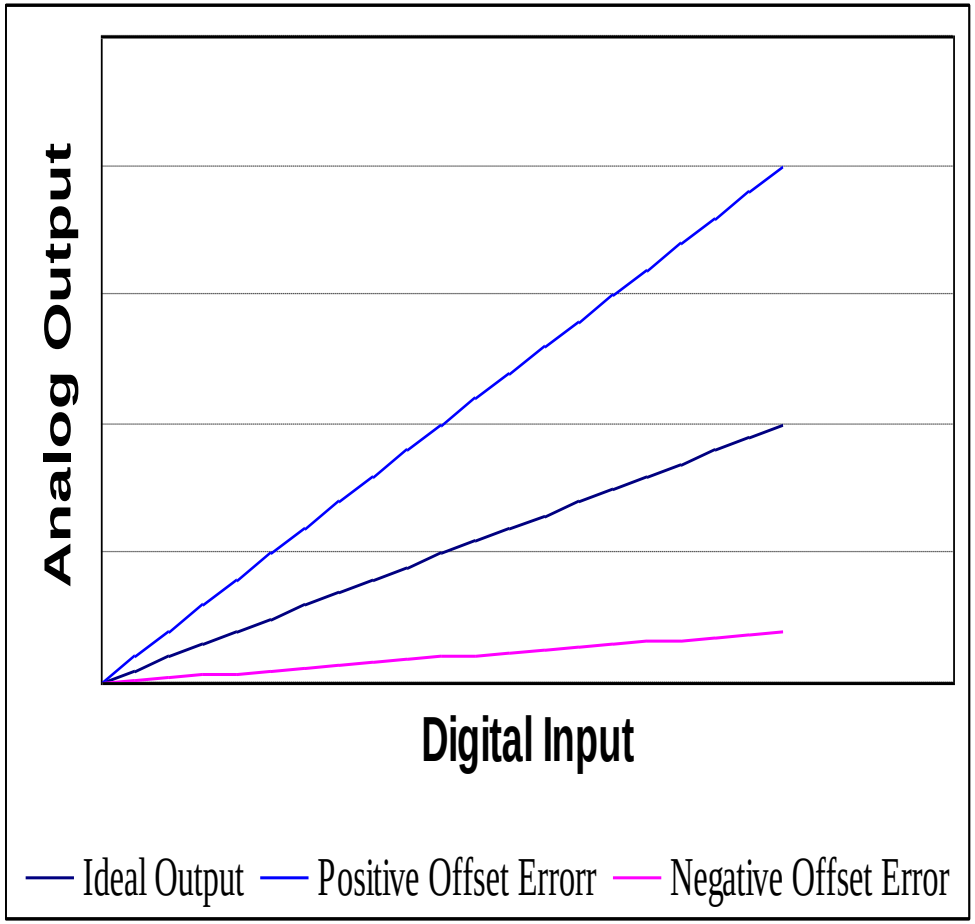


Types of Errors Associated with DACs

1. Gain
2. Offset
3. Full Scale
4. Resolution
5. Non-Linearity
6. Non-Monotonic
7. Settling Time and Overshoot

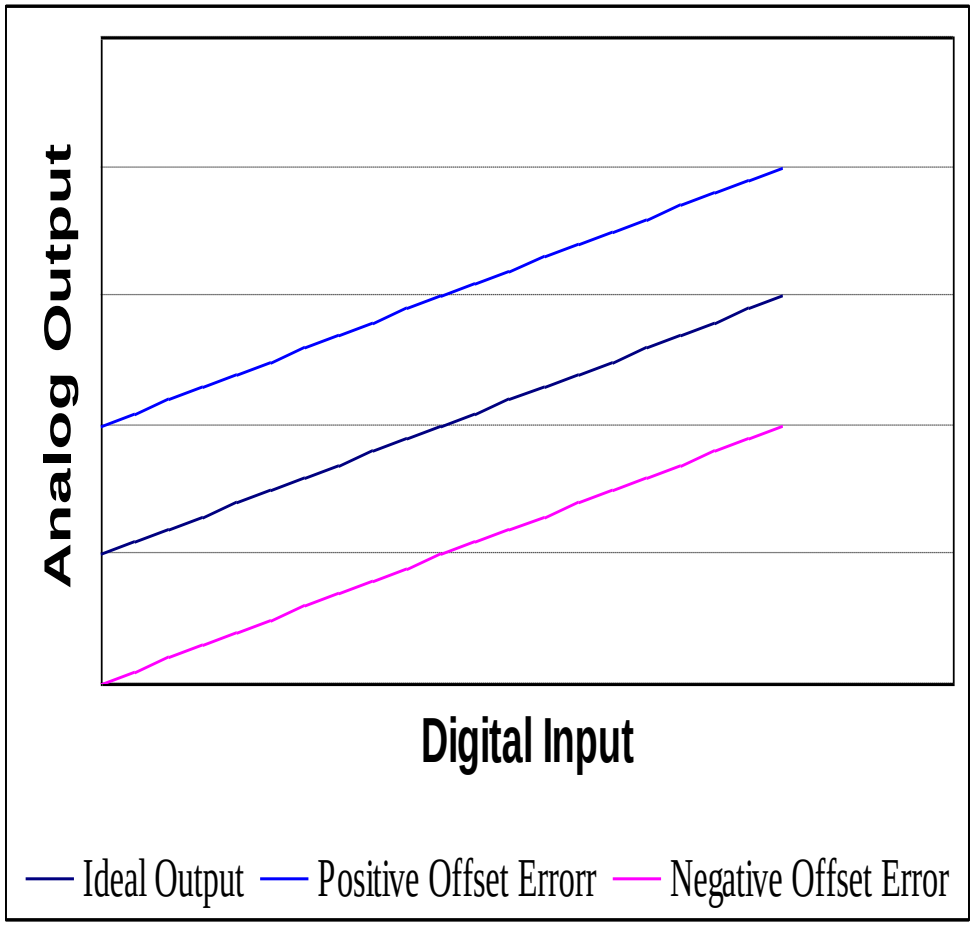
1. Gain Error

Occurs when the slope of the actual output deviates from the ideal output



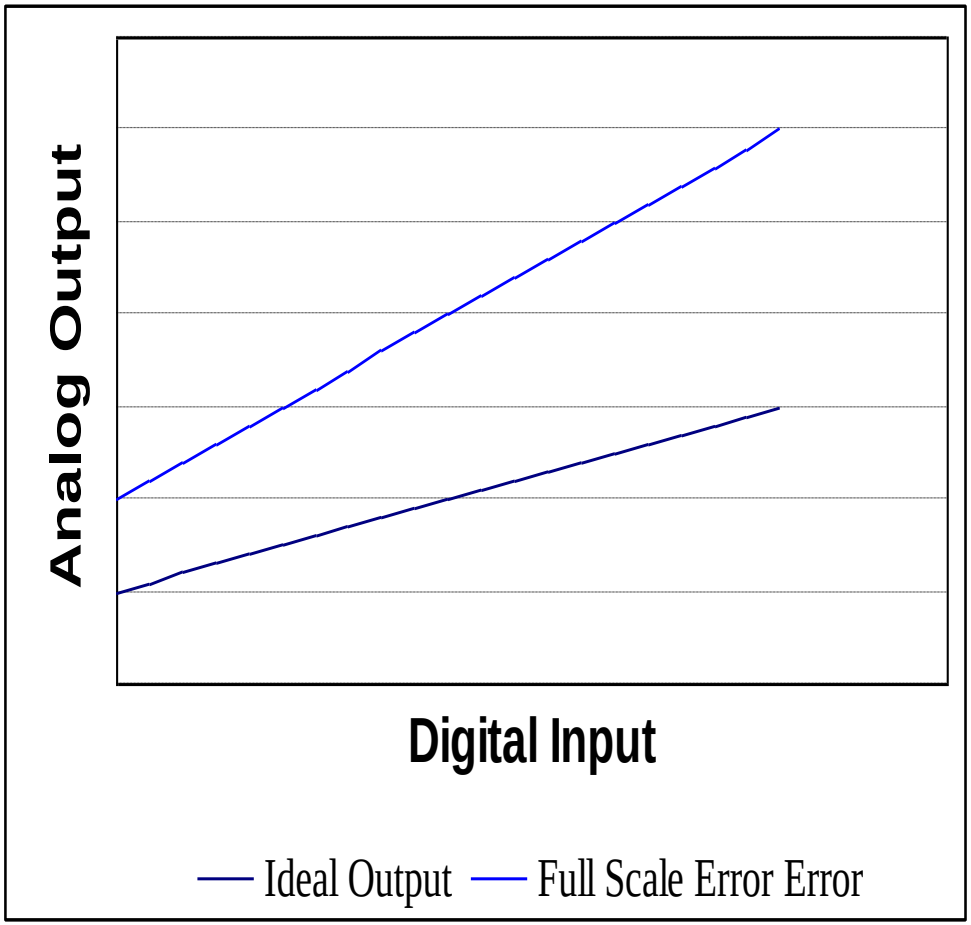
2. Offset Error

Occurs when there is a constant offset between the actual output and the ideal output



3. Full Scale Error

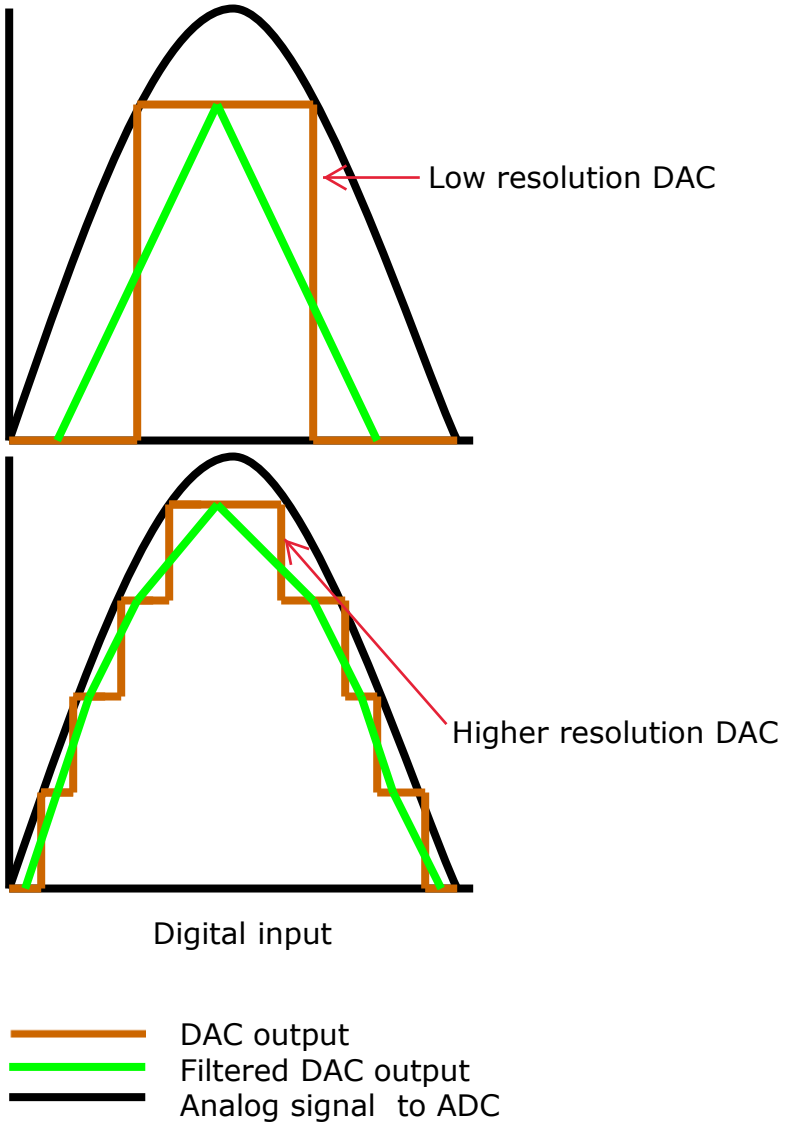
Occurs when the actual signal has **both gain** and **offset** errors



4. Resolution Error

Poor representation of ideal output due to poor resolution

Size of voltage divisions affect the resolution



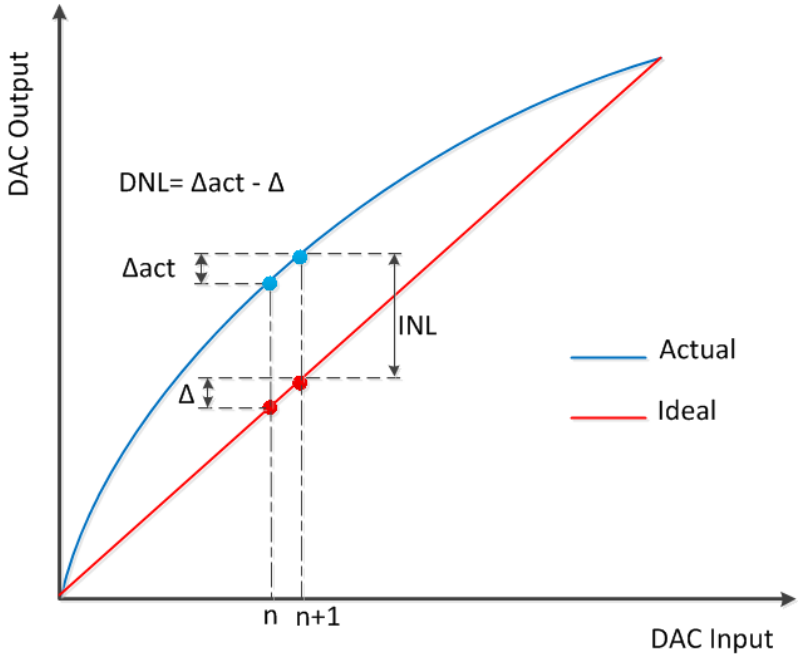
5. Non-Linearity Error

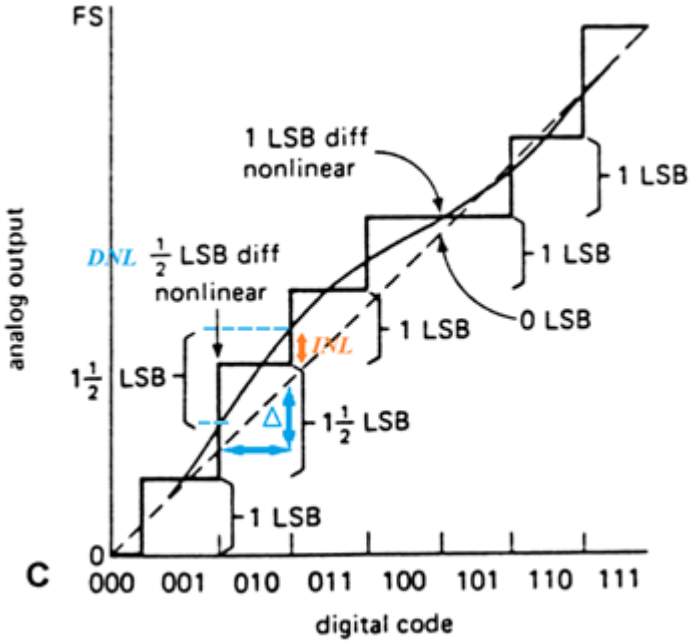
Occurs when analog output of signal is non-linear

Two Types

5.1 Differential (DNL) – analog step-sizes changes with increasing digital input (measure of largest deviation; between **successive bits**

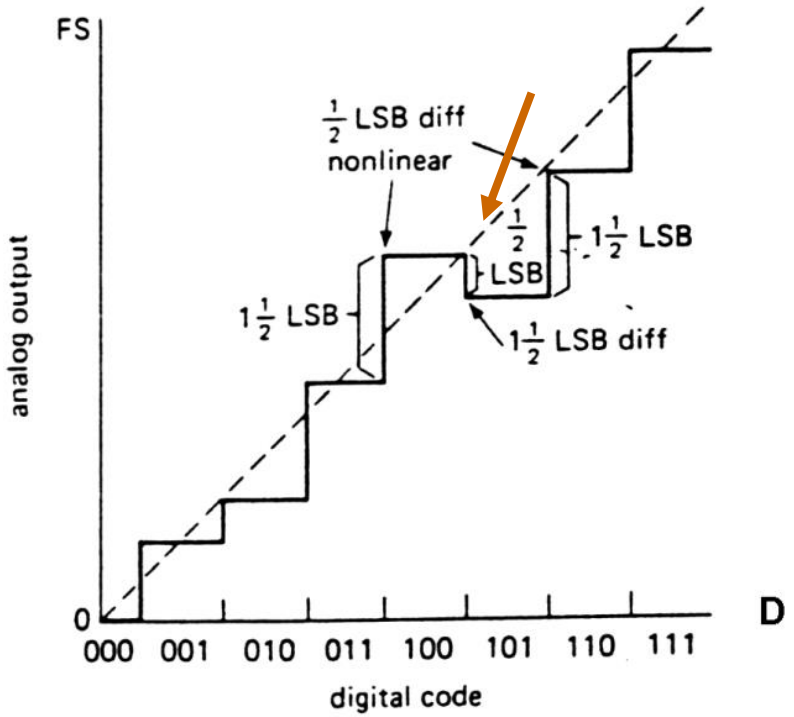
5.2 Integral (INL) – amount of deviation from a straight line after offset and gain errors removed; on **concurrent bits**





Non-Monotonic Error

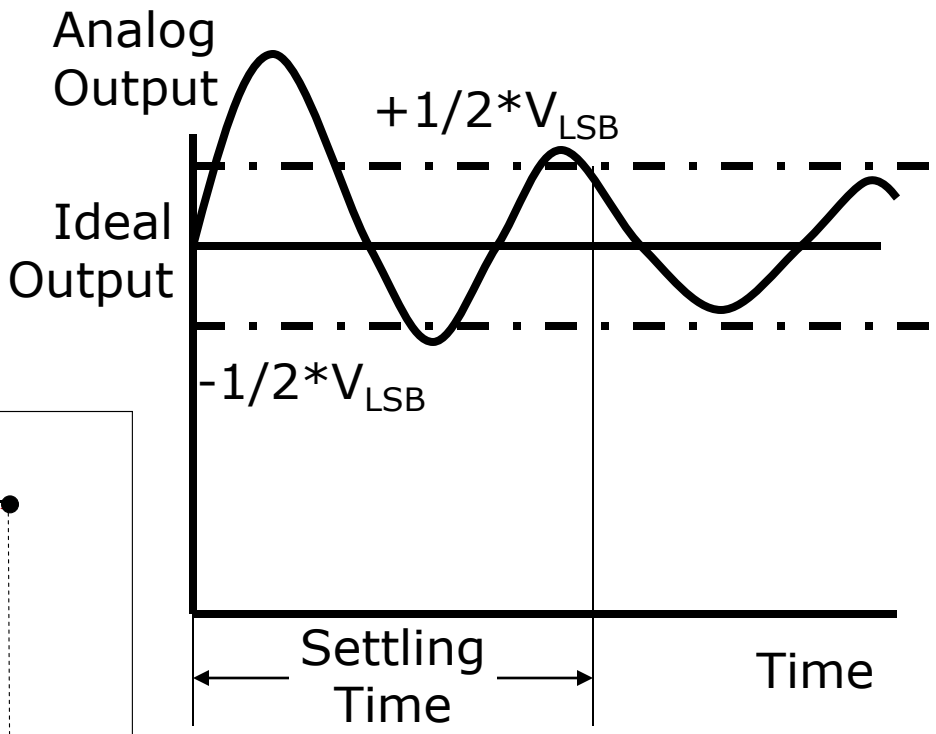
Occurs when an increase in digital input results in a decrease in the analog output

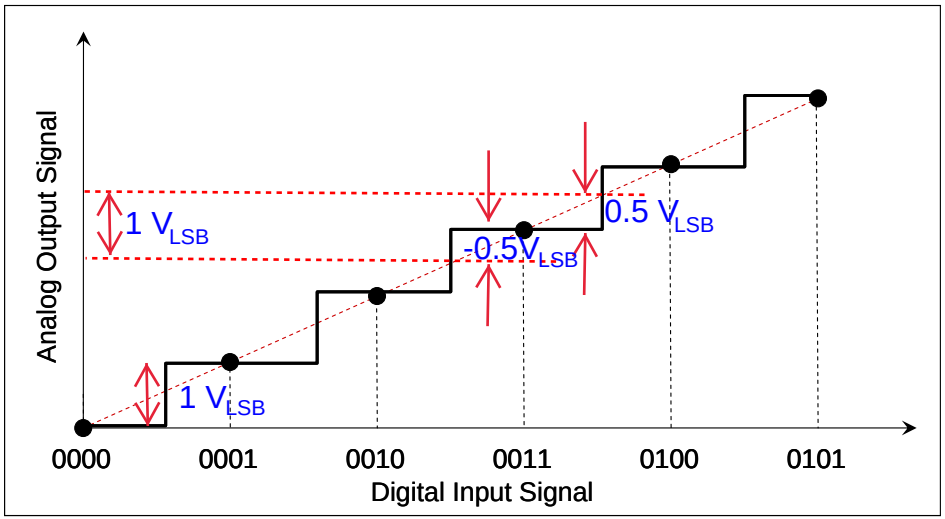


Settling Time and Overshoot Error

Settling Time – time required for the output to fall with in +/- ½ V_{LSB}

Overshoot – occurs when analog output overshoots the ideal output





Applications

Digital Motor Control

Computer Printers

Sound Equipment (e.g. CD/MP3 Players, etc.)

Electronic Cruise Control

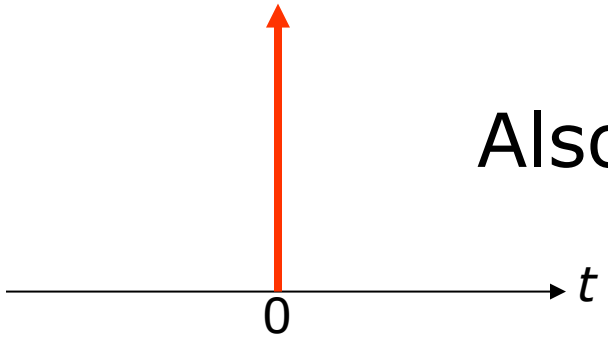
Digital Thermostat

Appendix

Unit Impulse (Dirac Delta) Function

Dirac Delta Function

$$\delta(t) = \begin{cases} 0 & t \neq 0 \\ \infty & t = 0 \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t)dt = 1$$



Also called *unit impulse function*.

Generalized Function

The value of delta function can also be defined in the sense of generalized function:

$$\int_{-\infty}^{\infty} \delta(t)\phi(t)dt = \phi(0)$$

$\phi(t)$: Test Function

- We shall never talk about the value of $\delta(t)$.
- Instead, we talk about the values of integrals involving $\delta(t)$.

Properties of Unit Impulse Function

$$\int_{-\infty}^{\infty} \delta(t - t_0)\phi(t)dt = \phi(t_0)$$

(proof)

Write t as $t + t_0$

$$\int_{-\infty}^{\infty} \delta(t - t_0)\phi(t)dt = \int_{-\infty}^{\infty} \delta(t)\phi(t + t_0)dt$$

$$= \phi(t_0)$$

Properties of Unit Impulse Function

$$\int_{-\infty}^{\infty} \delta(at)\phi(t)dt = \frac{1}{|a|}\phi(0)$$

(proof) Write t as t/a

Consider $a>0$; $d(t/a) = (-1/a)dt$

$$\int_{-\infty}^{\infty} \delta(at)\phi(t)dt$$

$$= \frac{1}{a}\int_{-\infty}^{\infty} \delta(t)\phi\left(\frac{t}{a}\right)dt$$

$$= \frac{1}{|a|}\phi(0)$$

Consider $a<0$; $d(t/a) = (-1/a)dt$

$$\int_{-\infty}^{\infty} \delta(at)\phi(t)dt$$

$$= -\frac{1}{a}\int_{-\infty}^{\infty} \delta(t)\phi\left(\frac{t}{a}\right)dt$$

$$= \frac{1}{|a|}\phi(0)$$

Properties of Unit Impulse Function

$$f(t)\delta(t) = f(0)\delta(t)$$

(proof)

$$\int_{-\infty}^{\infty} [f(t)\delta(t)]\phi(t)dt = \int_{-\infty}^{\infty} \delta(t)[f(t)\phi(t)]dt$$

$$= f(0)\phi(0)$$

$$= f(0)\int_{-\infty}^{\infty} \delta(t)\phi(t)dt$$

$$= \int_{-\infty}^{\infty} [f(0)\delta(t)]\phi(t)dt$$

Properties of Unit Impulse Function

$$f(t)\delta(t) = f(0)\delta(t)$$

$$\delta(at) = \frac{1}{|a|}\delta(t)$$

Pf)

$$\int_{-\infty}^{\infty} \underbrace{\delta(at)}\phi(t)dt = \frac{1}{|a|}\phi(0) = \frac{1}{|a|}\int_{-\infty}^{\infty}\delta(t)\phi(t)dt$$

$$= \int_{-\infty}^{\infty} \underbrace{\frac{1}{|a|}}\delta(t)\phi(t)dt$$

Properties of Unit Impulse Function

$$f(t)\delta(t) = f(0)\delta(t)$$

$$t\delta(t) = 0$$

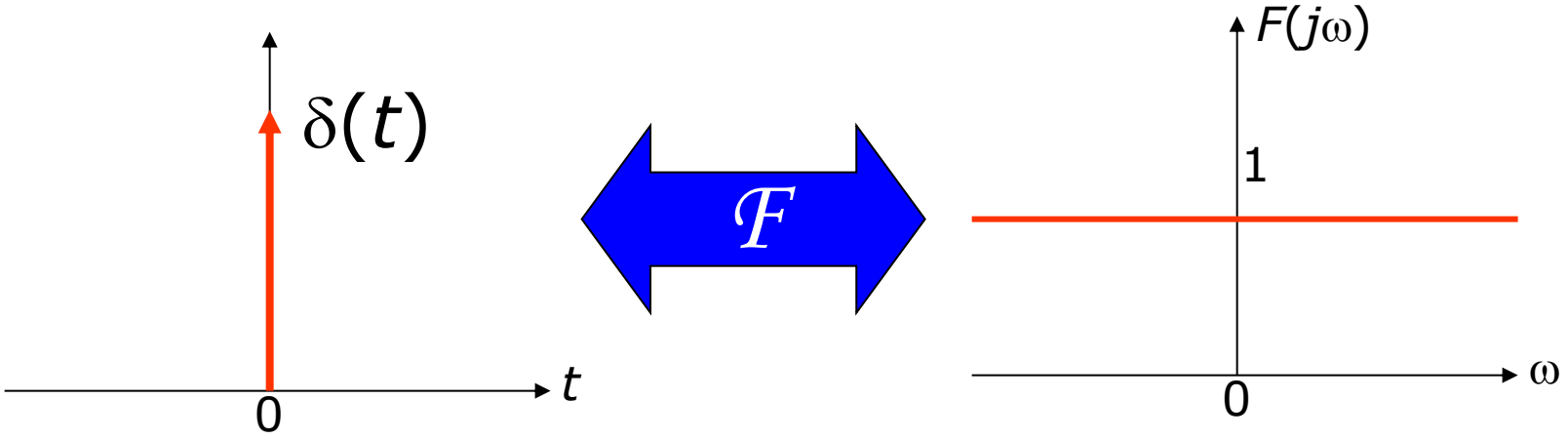
$$\delta(at) = \frac{1}{|a|}\delta(t)$$

$$\delta(-t) = \delta(t)$$

Fourier Transform for $\delta(t)$

$$\delta(t) \overset{\mathcal{F}}{\longleftrightarrow} 1$$

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = e^{-j\omega t}\Big|_{t=0} = 1$$



Fourier Transform for $\delta(t)$

Show that $$\delta(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{j\omega t} d\omega$$

$$\delta(t) = \mathcal{F}^{-1}[1] = \frac{1}{2\pi}\int_{-\infty}^{\infty} 1 e^{j\omega t} d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{j\omega t} d\omega$$

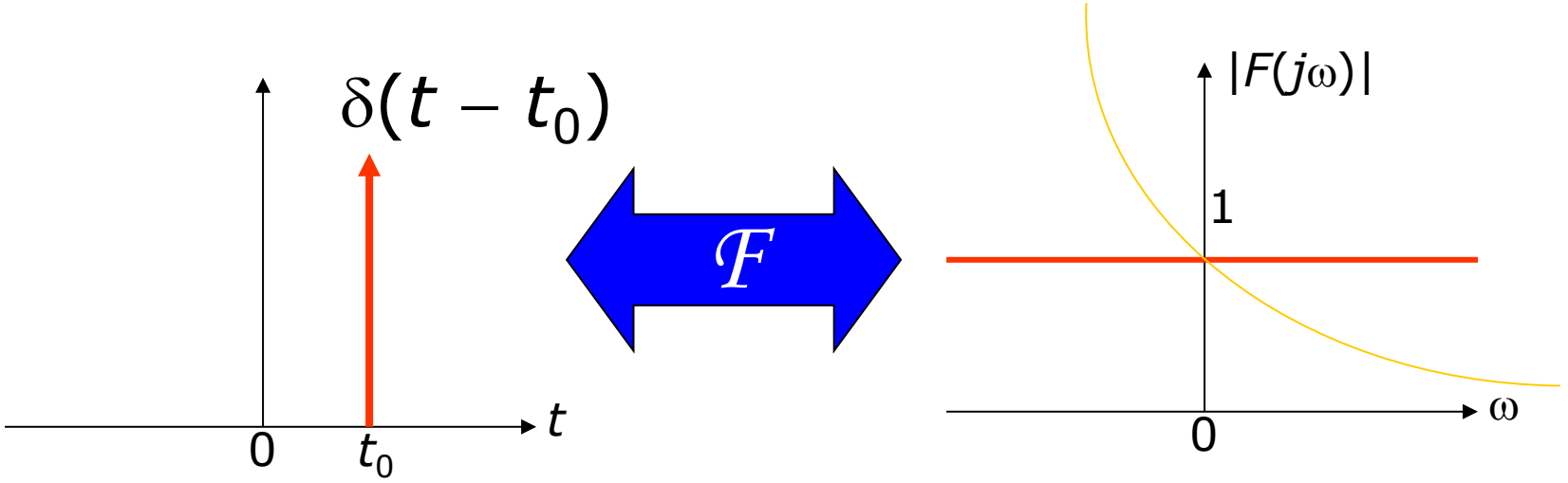
The integration $\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{j\omega t} d\omega$ converges to $\delta(t)$ in the sense of generalized function.

Shifted Impulse Function

$$\delta(t-t_0) \overset{\mathcal{F}}{\longleftrightarrow} e^{-j\omega t_0}$$

(proof) $$F[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$$

Use the fact $\mathcal{F}[f(t-t_0)] = F(j\omega)e^{-j\omega t_0}$



Find the FT of a Periodic Function

We can express a periodic function $f(t)$ as:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}, \qquad \omega_0 = \frac{2\pi}{T}$$

$$F(j\omega) = \mathcal{F}[f(t)] = \mathcal{F}\left[\sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}\right] = \sum_{n=-\infty}^{\infty} c_n \mathcal{F}[e^{jn\omega_0 t}]$$

$$= \sum_{n=-\infty}^{\infty} c_n 2\pi\delta(\omega - n\omega_0) = 2\pi \sum_{n=-\infty}^{\infty} c_n \delta(\omega - n\omega_0)$$

$$C_n = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)e^{-jn\omega_0 t}dt$$

Find the FT of a Periodic Function

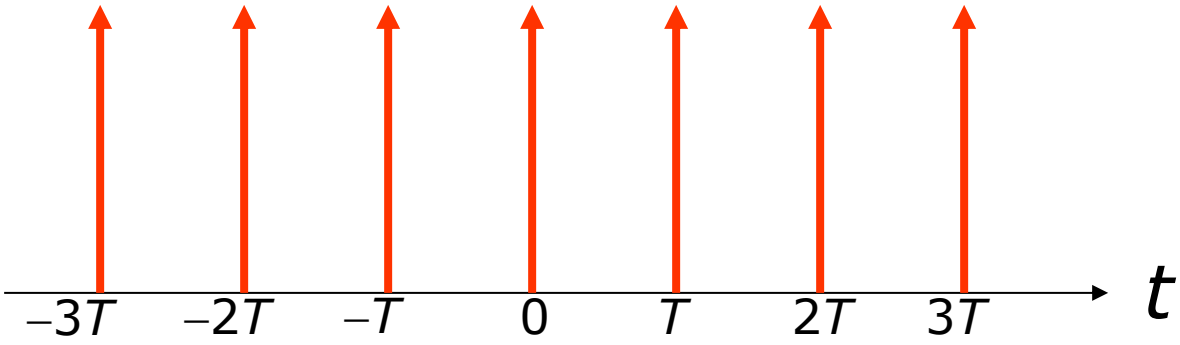
We can express a periodic function $f(t)$ as:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}, \qquad \omega_0 = \frac{2\pi}{T}$$

$$F(j\omega) = 2\pi \sum_{n=-\infty}^{\infty} c_n \delta(\omega - n\omega_0)$$

The FT of a periodic function consists of a sequence of equidistant impulses located at the harmonic frequencies of the function.

Example: Impulse Train

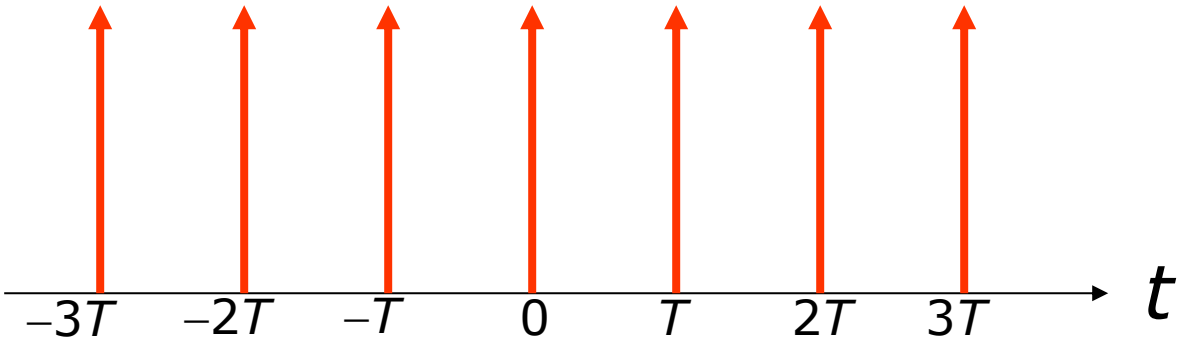


$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

Find the FT of the impulse train.

Example: Impulse Train

Find the FT of the impulse train.



$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$\delta_T(t) = \frac{1}{T} \sum_{n=-\infty}^{\infty} e^{jn\omega_0 t}$$

c_n

Example:Impulse Train

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$\delta_T(t) = \frac{1}{T} \sum_{n=-\infty}^{\infty} e^{jn\omega_0 t}$$

c_n

(proof) from the complex exponential Fourier series:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}, \qquad \omega_0 = \frac{2\pi}{T}$$

$$C_n = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) e^{-jn\omega_0 t} dt$$

$f(t)$	$F(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$

$$F(j\omega) = 2\pi \sum_{n=-\infty}^{\infty} c_n \delta(\omega - n\omega_0)$$

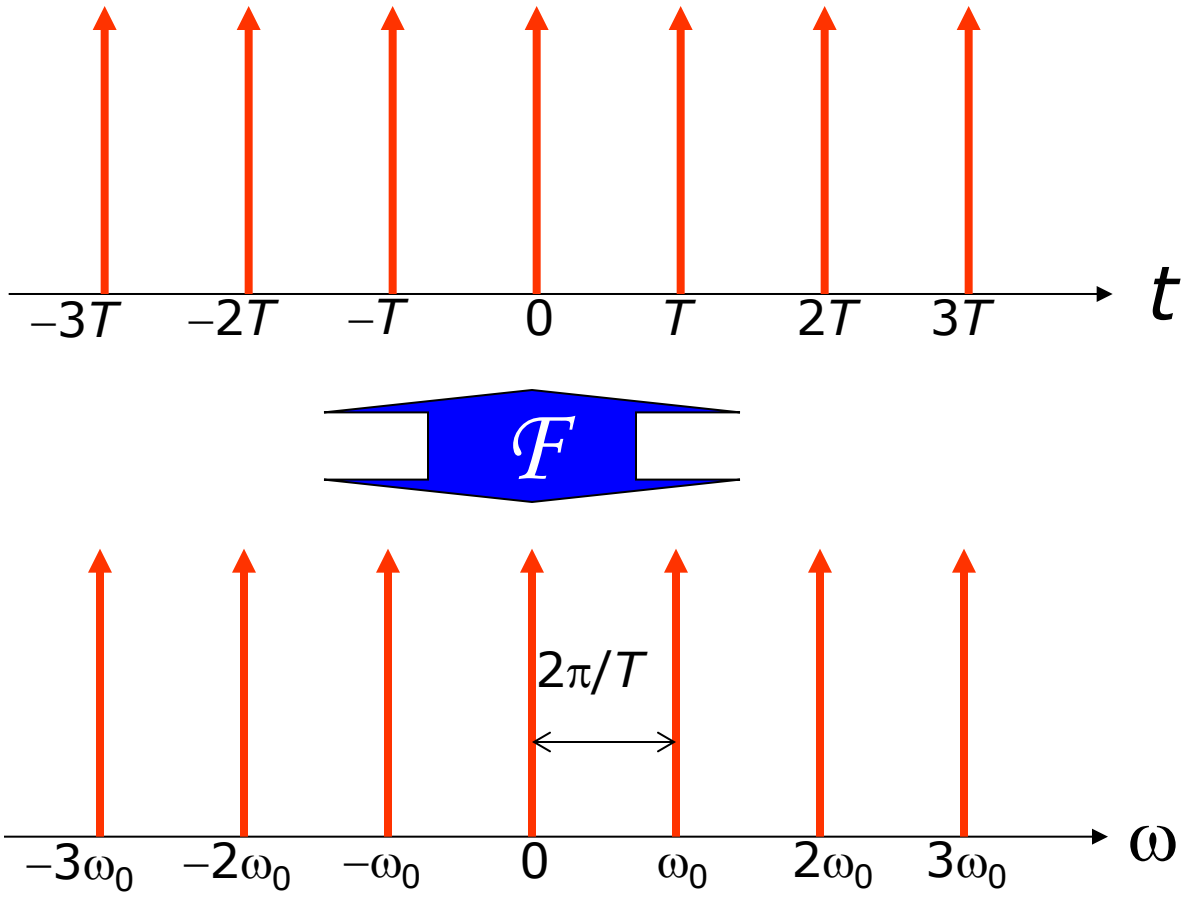
$$\mathcal{F}[\delta_T(t)] = \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$$

ω_0

Example:

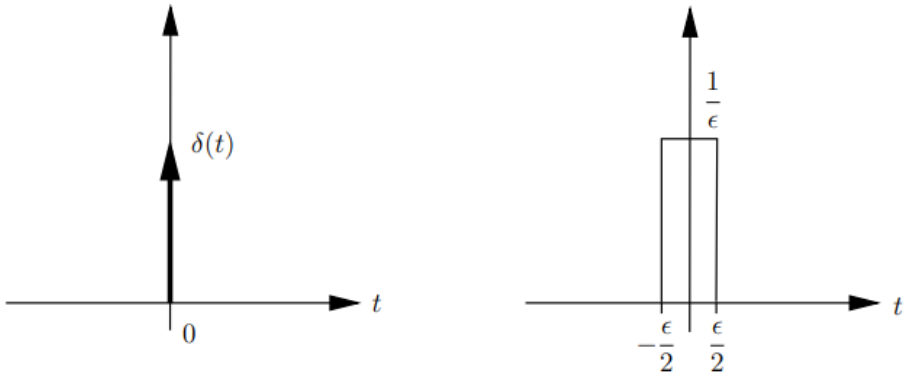
Impulse Train

$$\mathcal{F}[\delta_T(t)] = \underbrace{\frac{2\pi}{T}}_{\omega_0} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$$

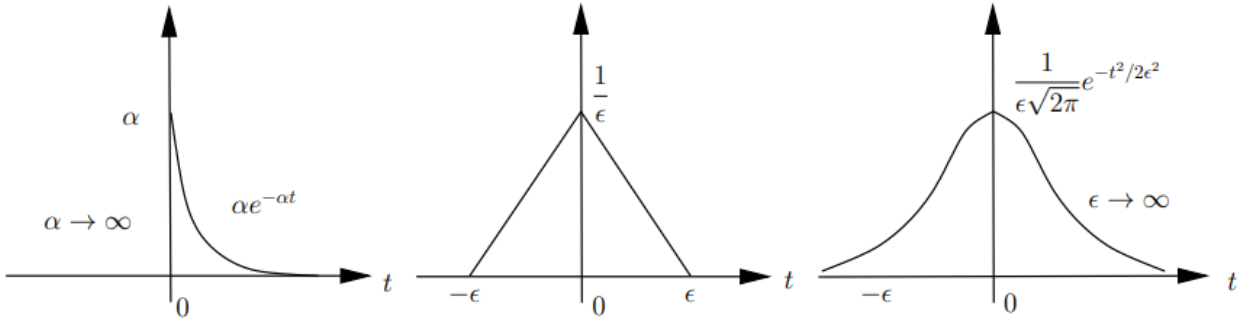


Unit impulse and its approximation

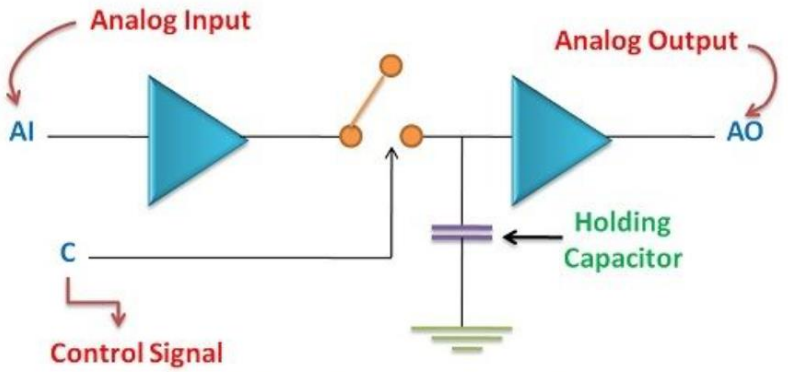
The Unit Impulse Function $\delta(t)$



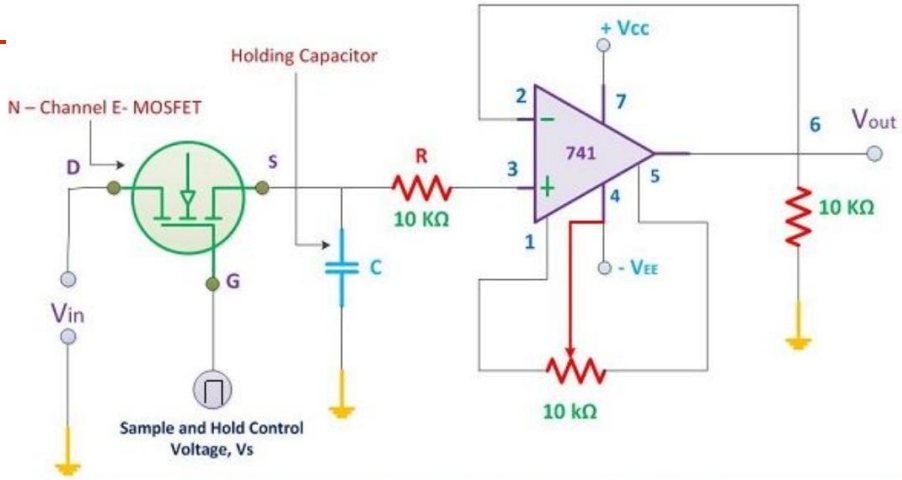
A unit impulse and its approximation.



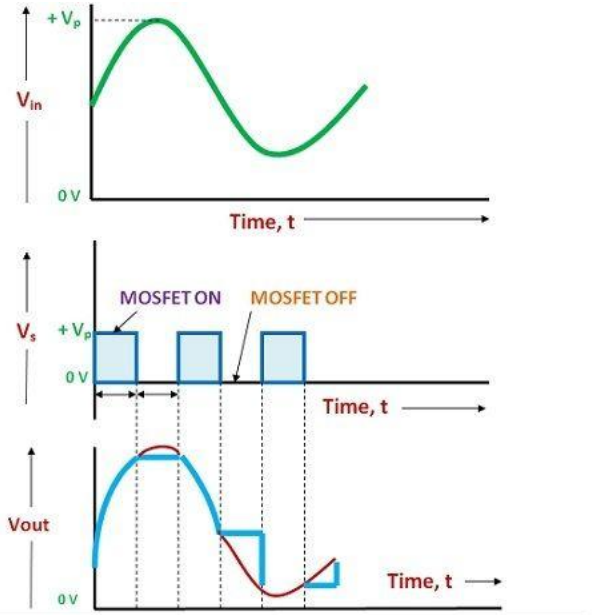
Other possible approximations to a unit impulse.



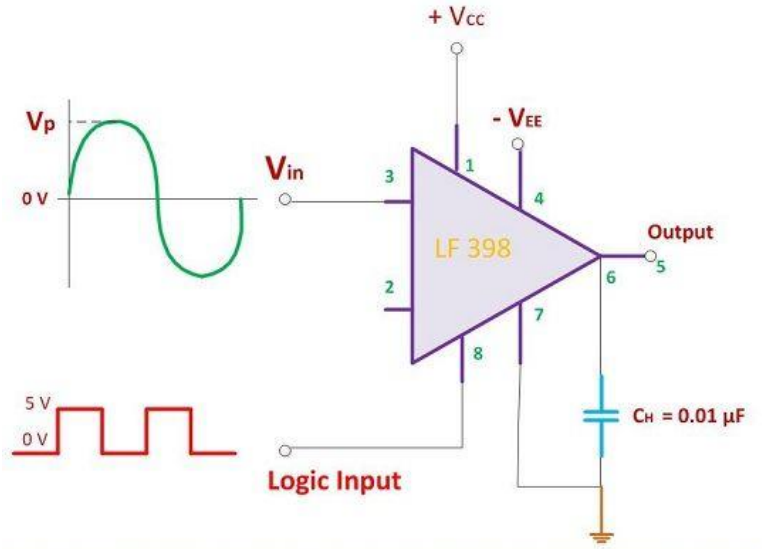
Circuit diagram of Sample and Hold Circuit



Practical Circuit of Sample and Hold Circuit



Input and Output Waveforms of Practical Sample and Hold Circuit



Typical Connection Diagram of Sample and Hold Circuit

Electronics Coach

Electronics Coach